

RESEARCH ARTICLE

Heterogeneous fusion of biometric and deep physiological features for accurate porcine cough recognition

Buyu Wang^{1,2*}, Jingwei Qi^{2,3}, Xiaoping An^{2,3}, Yuan Wang^{2,3}

1 College of Computer and Information Engineering, Inner Mongolia Agricultural University, Hohhot, Inner Mongolia, China, **2** Key Laboratory of Smart Animal Husbandry at Universities of Inner Mongolia Autonomous Region, Inner Mongolia Agricultural University, Inner Mongolia, China, **3** College of Animal Science, Inner Mongolia Agricultural University, Hohhot, Inner Mongolia, China

* bywang08@126.com

Abstract

Accurate identification of porcine cough plays a vital role in comprehensive respiratory health monitoring and diagnosis of pigs. It serves as a fundamental prerequisite for stress-free animal health management, reducing pig mortality rates, and improving the economic efficiency of the farming industry. Creating a representative multi-source signal signature for porcine cough is a crucial step toward automating its identification. To this end, a feature fusion method that combines the biological features extracted from the acoustic source segment with the deep physiological features derived from thermal source images is proposed in the paper. First, acoustic features from various domains are extracted from the sound source signals. To determine the most effective combination of sound source features, an SVM-based recursive feature elimination cross-validation algorithm (SVM-RFECV) is employed. Second, a shallow convolutional neural network (named ThermographicNet) is constructed to extract deep physiological features from the thermal source images. Finally, the two heterogeneous features are integrated at an early stage and input into a support vector machine (SVM) for porcine cough recognition. Through rigorous experimentation, the performance of the proposed fusion approach is evaluated, achieving an impressive accuracy of 98.79% in recognizing porcine cough. These results further underscore the effectiveness of combining acoustic source features with heterogeneous deep thermal source features, thereby establishing a robust feature representation for porcine cough recognition.

Introduction

Non-contact, stress-free health behavior detection in pigs is essential for smart farming, and pig interaction behavior recognition is essential for early disease diagnosis [1]. Cough serves as an early sign of respiratory disease in pig houses, making its monitoring essential for the early detection and treatment of such conditions [2,3]. However, relying on manual detection by resident veterinarians leads to delayed response times, substantial costs, and subjective results [4,5]. To address these challenges, many researchers have explored non-contact automatic

研究论文

基于生物特征与深度生理特征的异构融合实现精准猪咳嗽识别

购买王^{1,2*}, 齐景伟^{2,3}, 安晓平^{2,3}, 王元^{2,3}

1 计算机与信息工程学院, 内蒙古农业大学, 呼和浩特, 内蒙古, 中国, **2** 内蒙古自治区高校智能畜牧业重点实验室, 内蒙古农业大学, 内蒙古, 中国, **3** 动物科学学院, 内蒙古农业大学, 呼和浩特, 内蒙古, 中国

* bywang08@126.com

摘要

猪咳嗽的准确识别在综合呼吸道健康监测和猪病诊断中发挥着至关重要的作用。它是实现无压力动物健康管理、降低猪死亡率并提高畜牧业经济效益的基础前提。为自动识别猪咳嗽, 创建具有代表性的多源信号特征是关键步骤。为此, 本文提出了一种融合方法, 将声源段提取的生物特征与热源图像派生的深度生理特征相结合。首先, 从声源信号中提取不同领域的声学特征。为确定声源特征的最优组合, 采用了基于SVM的递归特征消除交叉验证算法 (SVM-RFECV)。其次, 构建了浅层卷积神经网络 (命名为 ThermographicNet), 用于从热源图像中提取深度生理特征。最后, 将这两种异构特征在早期阶段融合, 并输入支持向量机 (SVM) 进行猪咳嗽识别。通过严格实验, 评估了所提融合方法性能, 在猪咳嗽识别中取得了98.79%的惊人准确率。这些结果进一步证实了结合声源特征与异构深度热源特征的有效性, 从而为猪咳嗽识别建立了稳健的特征表示。

引言

猪的非接触式、无压力健康行为检测对于智慧农业至关重要, 而猪交互行为识别也至关重要

用于早期疾病诊断 [1]。咳嗽是猪舍中呼吸系统疾病的早期信号, 因此对其进行监测对于早期发现和治疗此类疾病至关重要 [2,3]。然而, 依赖驻场兽医进行人工检测会导致响应时间延迟、成本高昂且结果主观[4,5]。为了应对这些挑战, 许多研究人员已探索非接触式自动

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Data Availability Statement: Data and Code Availability Statement: Code Availability: The code supporting the findings of our study has been written in Python and is openly available on GitHub at <https://github.com/lwdmkzl/ThermalAndCoughHeterogeneousFusion>. Data Availability: According to the Livestock Cloud Platform data management requirements, our data cannot be shared directly. Our data comes from the Livestock Cloud Platform, we are licensed to use the heterogeneous fusion dataset and download the corresponding data as required. all who want to use the data must be a credentialed user and sign

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the data use agreement for the project from <http://yxm.imau.edu.cn/datalab/> website.

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recognition techniques for identifying pig cough, with a specific focus on pig cough sound recognition [6,7]. In addition, efforts have been made to improve the accuracy and robustness of porcine cough recognition by using infrared cameras and other sensors to capture physiological responses during coughing or observable external features associated with pig physiology [8,9]. These studies mainly focus on porcine cough physiology recognition.

Cough sound recognition has emerged as a low-cost, convenient, efficient, and non-invasive method for monitoring pig health. To enhance classification performance and robustness, researchers have explored various techniques tailored to different environments and application scenarios. These techniques have shown promising results in the field of cough sound recognition. Specifically, one approach involves utilizing machine learning algorithms to extract representative sound features from the time domain, frequency domain, and Mel frequency cepstrum domain. These features are then fed into a classifier for classification [10,11]. Another approach leverages convolutional neural networks to extract deep features from sound spectrograms, enabling effective classification [12,13]. A third approach combines the strengths of the previous two methods through multi-feature fusion classification [14,15]. These three approaches have demonstrated excellent performance in the task of cough sound recognition. However, enhancing representative acoustic features remains a bottleneck challenge, especially in field situations where homogeneous acoustic features make it difficult to achieve optimal classification results, whether using early fusion-feature fusion or late fusion-classifier fusion [16,17]. To overcome this challenge, researchers aim to break the performance bottleneck of cough sound classification by fusing heterogeneous features within the pig pen.

Cough physiological identification techniques leverage infrared cameras or other sensors to capture the physiological responses of pigs when they cough, along with observable external physiological features, to accurately identify coughs [18]. Among the existing methods, the development of physiological image sensors and the use of infrared cameras to capture physiological image data have been explored. Additionally, using Convolutional Neural Networks (CNN) to extract deep physiological features with higher accuracy from captured data with fewer parameters [19,20] has become the state of the arts for cough identification [21]. The development of physiological sensors, however, can be invasive and expensive [22]. As a result, the utilization of CNN to extract deep physiological features has gained traction due to its effectiveness. CNNs have been successfully applied to cough physiological recognition in two primary ways. One is an end-to-end approach for homogeneous physiological image classification tasks [23,24]. However, this approach faces the same performance bottleneck as homogeneous sound classification. The other approach extracts deep physiological features from the CNN and performs multimodal feature fusion with other heterogeneous features (e.g., biometric features). The fused feature vectors are then input into a lightweight classifier to complete the classification task [25,26]. In other words, the various improved CNN functions as a deep physiological feature extractor that can reduce computational costs to accelerate classification efficiency [27]. Moreover, the incorporation of additional dimensions of heterogeneous features helps address the bottleneck of classification performance improvement. Thus, adopting a custom shallow CNN architecture to extract deep physiological features and fusing them with acoustic features to feed a lightweight SVM classifier holds significant promise for porcine cough recognition.

This work aims to combine cough acoustic features with cough physiological features, taking into account the pathology of cough. The objective is to construct a robust and effective feature representation for porcine cough in pig housing by combining heterogeneous multidimensional fusion feature signals consisting of cough sound biosignal and cough body temperature infrared signal. This integration is expected to improve the performance of porcine cough recognition significantly. To achieve this, a three-stage approach is proposed. Firstly, acoustic features are extracted from the cough sound, and the SVM-RFECV algorithm is employed to

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识别猪咳嗽的技术, 特别关注猪咳嗽声音识别 [6,7]。此外, 通过使用红外相机和其他传感器捕捉咳嗽期间的生理反应或与猪生理相关的可观察外部特征, 已努力提高猪咳嗽识别的准确性和鲁棒性[8,9]。这些研究主要关注猪咳嗽生理识别。

咳嗽声音识别已成为一种低成本、便捷、高效且无创的监测猪健康的方法。为提升分类性能和鲁棒性, 研究人员探索了针对不同环境和应用场景的多种技术。这些技术在咳嗽声音识别领域已展现出良好效果。具体而言, 一种方法利用机器学习算法从时域、频域和梅尔频率倒谱域中提取代表性声音特征, 然后将这些特征输入分类器进行分类 [10,11]。另一种方法借助卷积神经网络从声音谱图中提取深度特征, 实现有效分类 [12,13]。第三种方法通过多特征融合分类结合前两种方法的优势 [14,15]。这三种方法在咳嗽声音识别任务中均表现出色。然而, 提升代表性声学特征仍是一大瓶颈挑战, 尤其是在猪圈等场环境下, 同质声学特征使得采用早期融合-特征融合或后期融合-分类器融合难以实现最佳分类效果, 无论使用哪种方法 [16,17]。为克服这一挑战, 研究人员旨在通过融合猪圈内的异构特征来打破咳嗽分类的性能瓶颈。

咳嗽生理识别技术利用红外相机或其他传感器捕捉猪咳嗽时的生理反应, 以及可观察的外部生理特征, 以准确识别咳嗽 [18]。在现有方法中, 生理图像传感器的发展以及使用红外相机捕获生理图像数据的探索已被研究。

此外, 使用卷积神经网络(CNN)以更少的参数从捕获数据中提取更高准确率率的深度生理特征[19,20]已成为咳嗽识别 [21]的当前技术。然而, 生理传感器的发展可能是侵入性的且昂贵的 [22]。因此, 利用CNN提取深度生理特征因其有效性而受到关注。CNN已成功应用于咳嗽生理识别的两种主要方式。一种是针对同质生理图像分类任务的端到端方法 [23,24]。然而, 这种方法面临与同质声音分类相同的性能瓶颈。另一种方法是从CNN中提取深度生理特征, 并将其与其他异构特征(例如生物特征)进行多模态特征融合。融合后的特征向量然后输入到轻量级分类器中以完成分类任务 [25,26]。换句话说, 各种改进的CNN充当深度生理特征提取器, 可以降低计算成本以加速分类效率 [27]。此外, 结合额外的异构特征维度有助于解决分类性能提升的瓶颈。因此, 采用定制浅层CNN架构提取深度生理特征, 并将其与声学特征融合以供轻量级SVM分类器使用, 对猪咳嗽识别具有重要意义。

本研究旨在结合咳嗽声学特征与咳嗽生理特征, 并考虑咳嗽的病理学。目标是构建一种鲁棒且有效的特征表示, 用于猪舍中猪咳嗽的识别, 该特征表示结合了由咳嗽声音生物信号和咳嗽体温红外信号组成的异构多维融合特征信号。这种集成有望显著提高猪咳嗽识别的性能。为此, 提出了一种三阶段方法。首先, 从咳嗽声音中提取声学特征, 并采用SVM-RFECV算法

select the optimal acoustic source biometric features. Secondly, a custom CNN is used to extract thermal source deep physiological features from infrared thermal images, and the feature vectors of the two fully connected layers are fused to obtain layer fusion deep physiological features. Lastly, the acoustic source biometric features are early fused with the thermal source deep physiological features and layer fusion deep physiological features, respectively. These fused features are then fed into a lightweight SVM classifier to complete the classification task.

In summary, this study makes the following contributions:

1. Proposal of a novel framework: This study introduces a novel framework that combines acoustic source features with deep physiological features for pig cough recognition.
2. Development of a feature selection method: To optimize the recognition process, a feature selection method is proposed to extract a representative set of acoustic features of pig coughs.
3. Construction of a CNN architecture: A carefully designed CNN architecture is constructed to extract deep physiological features from thermal images to enhance the recognition performance of pig coughs.
4. Comparative analysis with existing models: The proposed method outperforms existing CNN models in terms of recognition speed, model size, and classification performance.

The remainder of this paper is organized as follows. Materials Section provides the work related to the materials. In Methods Section, the methods involved in the experiments are comprehensively described. The results obtained from our experiments are presented in Experiments and Results Section. Discussion Section offers a discussion of the results. Finally, conclusions are drawn in Conclusions Section.

Materials

Housing and laboratory animals

The data used in this study were collected in a real farming environment in the isolation barn of a large commercial fattening pig farm in Hohhot, Inner Mongolia Autonomous Region, China. The fattening farm has a production scale of 100,000 hogs and raises three breeds, including Large White, Long White, and Duroc, and abnormally sick pigs are sent to the isolation barns for isolation. One of the isolation barns with a capacity of 218 pigs was selected for this experiment. The resident veterinarians isolated pigs of different breeds, age groups, and coughing conditions into the isolation barn was selected. The experiment lasted for 15 days, and the pigs in the isolation barn were constantly transferred in/out of the barn in a dynamic and real-time manner to ensure complete coverage of pigs with different conditions. The experimental isolation barn comprises three units, A, B, and C, each containing seven pens. A1 and C1 house the coughing pigs for the experiments (up to 14 pigs per pen), and A2, B1, B2, and C2 hose the recovering pigs for the experiments (occasional coughing or no coughing, up to 10 pigs per pen), with pens next to each other and interacting with each other in order to reproduce the real environment (coughing) in the production barn. The pens are located next to each other and interact, completely reproducing the real environment in a production barn (coughing pigs surrounded by non-coughing pigs, interacting with each other). The layout of the experimental pigs and pens is shown in [Fig 1](#).

Data acquisition and preprocessing

The data collection system uses a real production system deployed in the experimental pig house and adopts a cloud-edge-end architecture. The cloud terminal of this experiment

选择最优声源生物特征。其次，使用定制的CNN从红外热成像中提取热源深度生理特征，并将两个全连接层的特征向量融合以获得层融合深度生理特征。最后，声源生物特征分别与热源深度生理特征和层融合深度生理特征进行早期融合。这些融合特征随后被输入到轻量级SVM分类器中以完成分类任务。

总之，本研究做出了以下贡献：

1. 提出了一种新型框架：本研究引入了一种将声源特征与深度生理特征相结合的猪咳嗽识别框架。
2. 开发了一种特征选择方法：为了优化识别过程，提出了一种特征选择方法，用于提取猪咳嗽的代表性声学特征集。
3. 构建了一种CNN架构：构建了一种精心设计的CNN架构，用于从热图像中提取深度生理特征，以增强猪咳嗽的识别性能。

4. 与现有模型的对比分析：所提出的方法在识别速度、模型大小和分类性能方面优于现有的CNN模型。

本文的其余部分组织如下。材料部分提供了与材料相关的工作。在方法部分，全面描述了实验中涉及的方法。实验结果在实验与结果部分中呈现。讨论部分对结果进行了讨论。最后，结论部分总结了结论。

材料

住房和实验动物

本研究使用的数据是在中国内蒙古自治区呼和浩特市一家大型商业育肥猪场的隔离畜舍中收集的。该育肥场年生产规模为10万头猪，饲养大白、长白和杜洛克三个品种，异常生病的猪会被送往隔离畜舍隔离。本研究选择了其中一座可容纳218头猪的隔离畜舍进行实验。驻场兽医将不同品种、年龄组和咳嗽状况的猪隔离到

该隔离畜舍被选中。实验持续了15天，隔离畜舍中的猪以动态和实时的方式不断进出畜舍，以确保不同状况的猪都能被完全覆盖。实验隔离畜舍由A、B和C三个单元组成，每个单元包含七个畜栏。A1和C1畜栏饲养用于实验的咳嗽猪（每栏最多14头），A2、B1、B2和C2畜栏饲养用于实验的恢复猪（偶尔咳嗽或无咳嗽，每栏最多10头），相邻畜栏相互交互，以模拟生产畜舍中的真实环境（咳嗽）。畜栏相邻且相互交互，完全模拟了生产畜舍中的真实环境（咳嗽猪被无咳嗽猪包围，相互交互）。实验猪和畜栏的布局如图1所示[图1](#)。

数据采集与预处理

该数据采集系统采用部署在实验猪舍中的真实生产系统，并采用云边端架构。本实验的云终端

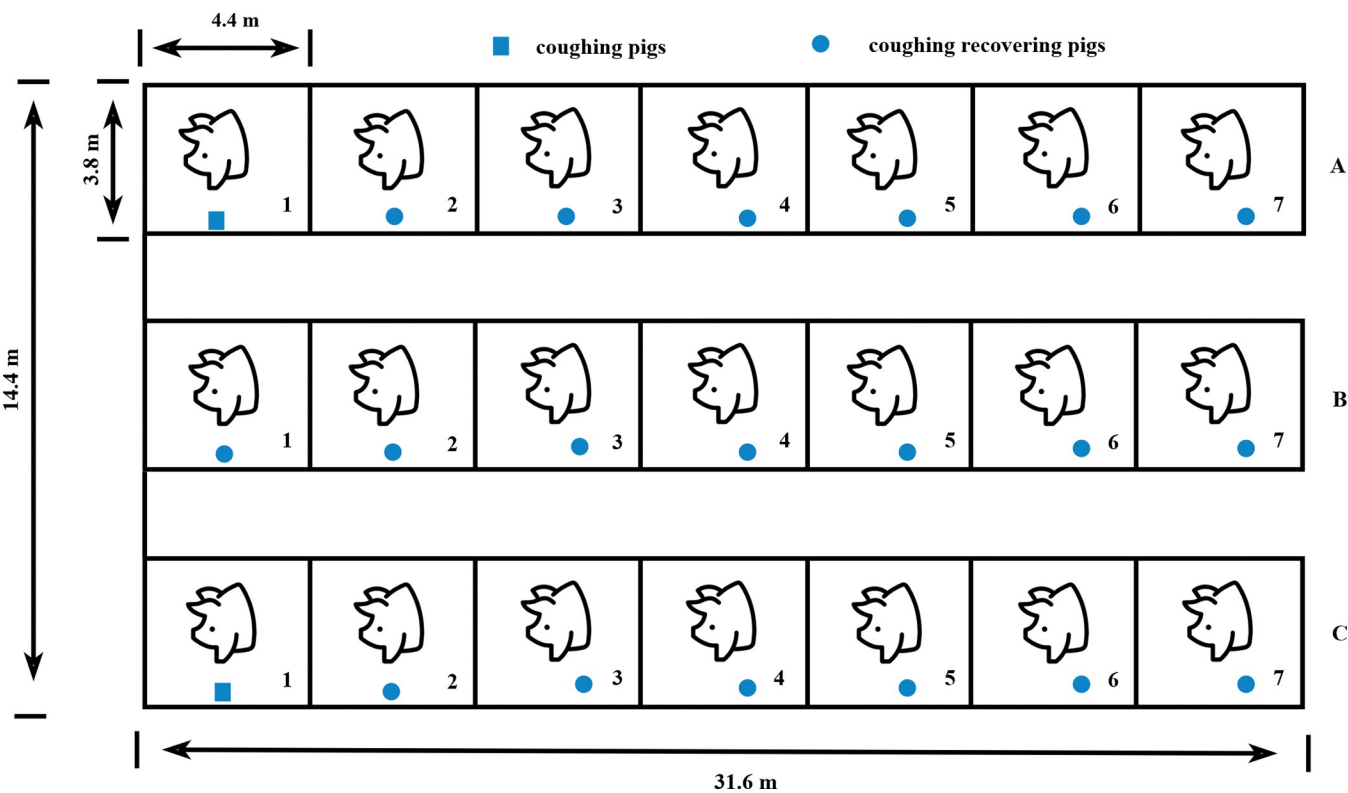


Fig 1. Layout of the experimental pig house.

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includes a pickup for collecting cough sounds and a camera for infrared data deployed in the pig house. The camera and pickup are consistently gathering audio and video data within the pig house, operating 24/7. The camera is a binocular dome camera (TB-1217A-3/PA, Hikvision, Hangzhou, China. Visible light: 25FPS, 4096Kbps, H.264. IR: 1280×720, 25fps, 4096Kbps, H.264 coding), the camera is fixed on a 2-meter-high wall on one side of the pigsty, with an adjustable viewing angle, which makes it easy to cover the whole pigsty. The pickup was an audio recorder (DS-2FP4121-OW-AI, Hikvision, Hangzhou, China, 48kHz, 64kbps, AAC). The cameras and pickups are connected with special audio/video adapter cables. In this way, the installation position of the equipment can be flexibly selected according to different pigsty environments, and the distance between the camera and the pickup can be flexibly adjusted. The edge server is a desktop PC server (OptiPlex 3070, Dell, USA) located within the pig farm’s server room, establishing a seamless connection with the cameras via Ethernet. Through the utilization of the FFMPEG software, the Edge server effectively captures real-time feeds from both the visible light channel (1920×1080) and thermal imaging channel (1280×720) using the RTSP protocol. Real-time acquisition to the server hard drive, respectively, saved as the acquisition start time as the file name of the mp4 file, i.e., visible light is named as “acquisition start time _ camera number.mp4” and thermal imaging is named as “acquisition start time _ camera number _ tf.mp4”. Importantly, each mp4 file contains audio data and generates new audio and video files every 10 minutes. To facilitate synchronized data transfer between the Edge server of the pig farm and the Storage server of the cloud animal husbandry platform [28], a robust Internet connection links the two. The Rsync file synchronization protocol plays a pivotal role in automatically and incrementally transferring audio and video files from the pig farm to the cloud animal husbandry platform. Upon successful

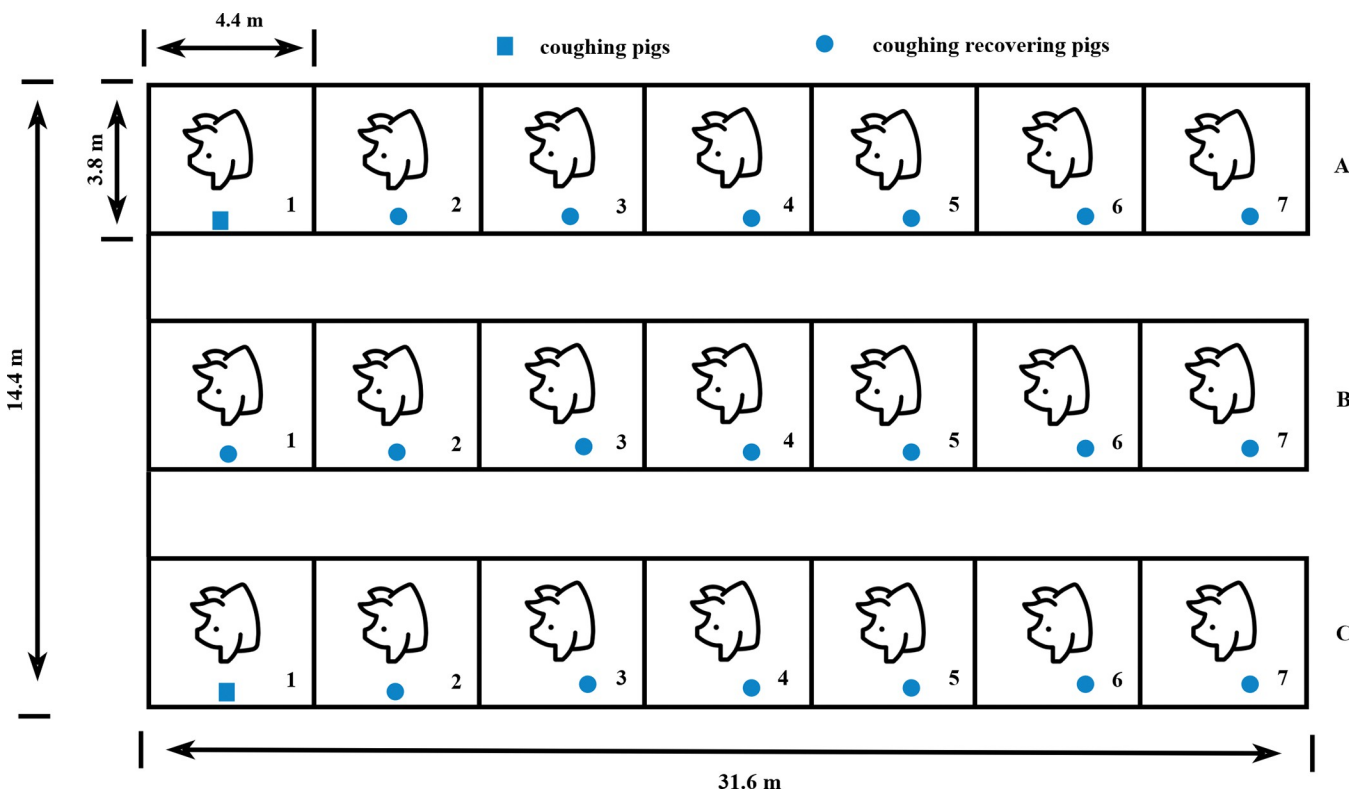


图1. 实验猪舍布局。

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包括用于收集咳嗽声音的拾音器和部署在猪舍中的红外数据摄像头。摄像头和拾音器始终在猪舍内采集音频和视频数据，24/7运行。该摄像头是一款双目球型摄像头（TB-1217A-3/PA，海康威视，杭州，中国。可见光：25FPS，4096Kbps，H.264。红外：1280×720，25fps，4096Kbps，H.264编码），摄像头固定在猪舍一侧2米高的墙上，具有可调节的视角，便于覆盖整个猪舍。拾音器是一款音频记录仪（DS-2FP4121-OW-AI，海康威视，杭州，中国，48kHz，64kbps，AAC）。摄像头和拾音器通过专用音视频适配线缆连接。这样，根据不同的猪舍环境，可以灵活选择设备的安装位置，并灵活调整摄像头与拾音器之间的距离。边缘服务器是一台台式PC服务器（OptiPlex 3070，戴尔，美国），位于猪场的服务器室内，通过以太网与摄像头建立无缝连接。通过利用FFMPEG软件，边缘服务器通过RTSP协议有效捕获来自可见光通道（1920×1080）和热成像通道（1280×720）的实时流。实时采集到服务器硬盘，分别以采集开始时间为文件名的mp4文件保存，即可见光命名为“采集开始时间_摄像头编号.mp4”，热成像命名为“采集开始时间_摄像头编号_tf.mp4”。重要的是，每个mp4文件包含音频数据，并每10分钟生成新的音视频文件。为便于猪场边缘服务器与云畜牧业平台存储服务器的同步数据传输，[28]，强大的互联网连接将两者连接起来。Rsync文件同步协议在自动增量地将猪场的音视频文件传输到云畜牧业平台方面发挥着关键作用。成功

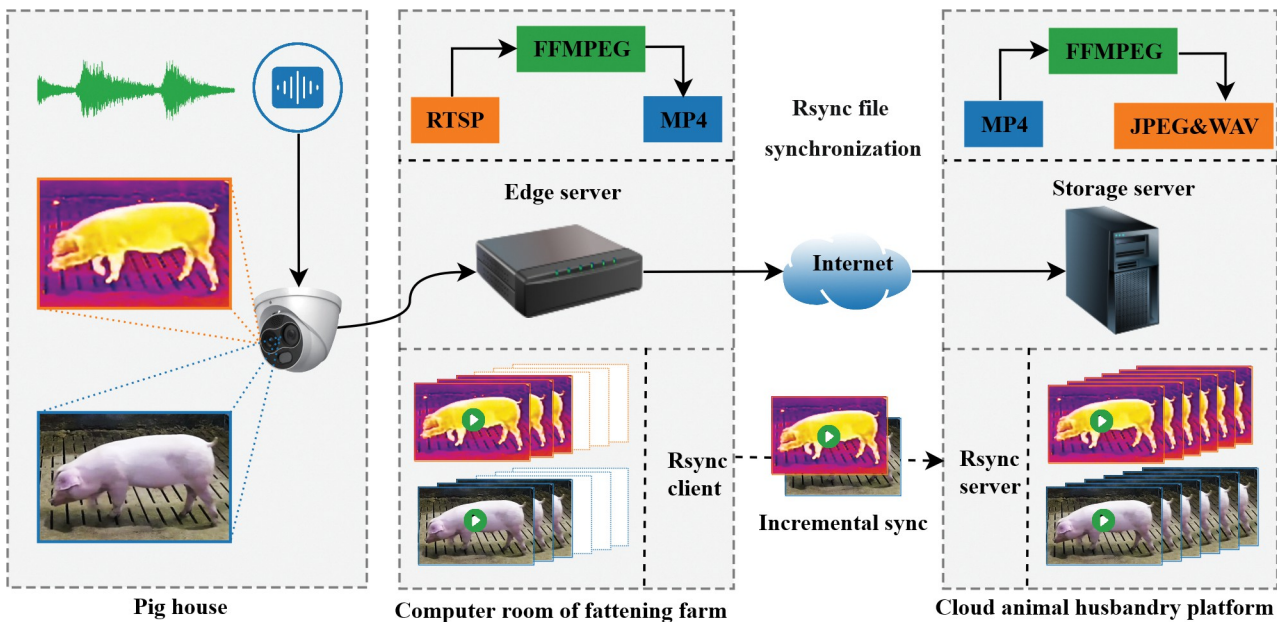


Fig 2. Multimodal data acquisition system based on cloud-edge-end architecture.

<https://doi.org/10.1371/journal.pone.0297655.g002>

synchronization, files are automatically deleted from the Edge server. The Storage server within the cloud animal husbandry platform employs FFMPEG software once again to extract audio from the received visible light audio and video files. These audio segments are then stored as a.wav file, adopting the corresponding video file's name format ("Acquisition Start Time_Camera No.wav" and "Original Audio and Video File Name.wav"). The data acquisition system used in this study is a real production data acquisition system with good end-to-end generalization capability. Finally, 2160 original data sets were collected over 15 days using a cloud-edge-end data acquisition system. Each set comprises a 10-minute audio WAV file and its corresponding 10-minute infrared video MP4 file. The architecture of the data acquisition system is shown in Fig 2.

For the 2160 sets of raw data files collected, the lens occlusion blurred frames, feeding, immunization, treatment, and other feeding activity-related frames in the raw data were eliminated, and finally, 135 hours of valid audio (WAV) and video files (MP4) were obtained. The visible video is in RGB mode during the day and converted to grayscale mode at night, which is used to help experts in data annotation. The image palette mode for the thermal imaging video was set to "Iron Red".

Leveraging the versatile Librosa tool [29], specific audio segments were meticulously extracted from the WAV files, applying predefined conditions (volume threshold $\geq 33\text{dB}$, length threshold $\geq 2\text{s}$). The nomenclature of these WAV files was crafted to incorporate the commencement timestamp, enhancing organizational clarity. Subsequently, the WAV file dataset underwent a clustering process utilizing the K-means algorithm. This algorithm, applied to the audio cough data, facilitated the initial identification of clusters associated with coughing events. To refine and classify these clusters, a 10th-order Butterworth filter with a cutoff frequency ranging from 100Hz to 16kHz was employed. The resultant one-dimensional filtered sound data was then transformed into two-dimensional Mel-Frequency Cepstral Coefficients (MFCC) data, enhancing the representation of acoustic features. Further refinement ensued through the application of the K-means algorithm, selectively isolating clusters relevant to coughing sounds. A collaborative effort involving subject matter experts, resident

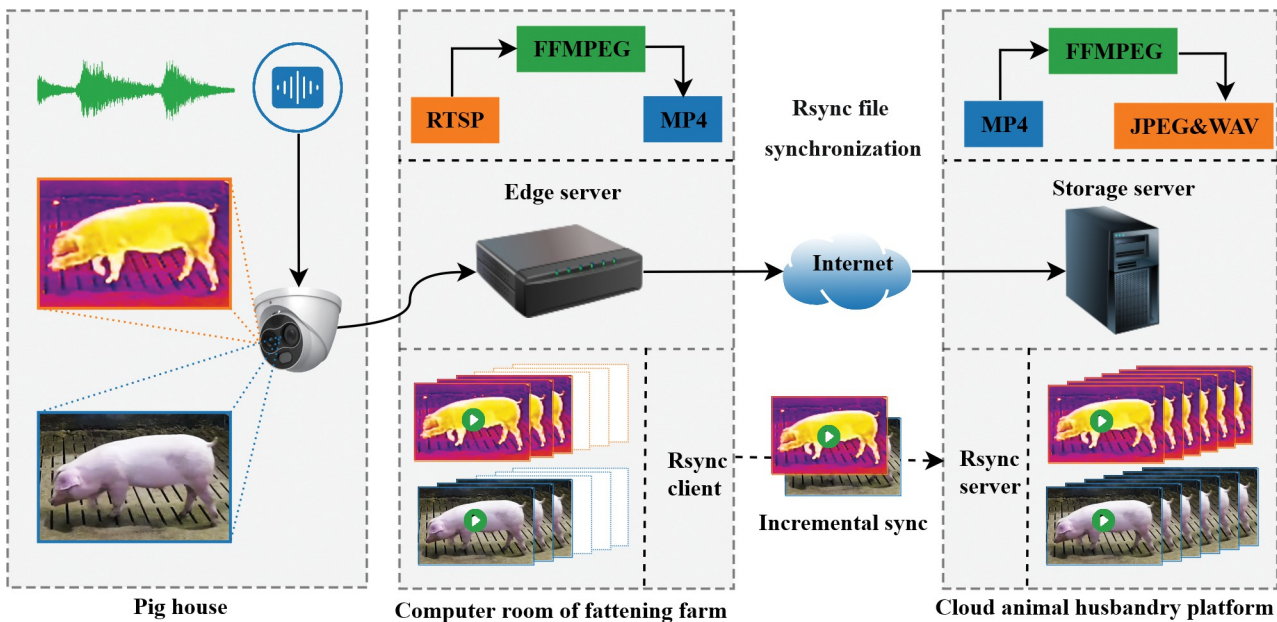


图2。基于云边端架构的多模态数据采集系统。

<https://doi.org/10.1371/journal.pone.0297655.g002>

同步时，文件会自动从边缘服务器删除。云畜牧业平台内的存储服务器再次使用 FFMPEG 软件，从接收到的可见光音频和视频文件中提取音频。这些音频片段随后被存储为.wav 文件，并采用相应视频文件的命名格式（“采集开始时间_相机编号.wav”和“原始音频和视频文件名.wav”）。本研究中使用的数据采集系统是一个具有良好端到端泛化能力的生产实际数据采集系统。最终，使用云-边-端数据采集系统在15天内收集了2160个原始数据集。每个数据集包含一个10分钟的音频WAV文件及其对应的10分钟红外视频MP4文件。数据采集系统的架构如图2所示。

对于收集的2160组原始数据文件，消除了原始数据中的镜头遮挡模糊帧、喂食、免疫、治疗以及其他与喂食活动相关的帧，最终获得了135小时的有效音频（WAV）和视频文件（MP4）。白天的可见视频以RGB模式显示，夜间转换为灰度模式，用于帮助专家进行数据标注。热成像视频的图像调色板模式设置为“铁红”。

利用功能强大的Librosa工具 [29]，从WAV文件中精细提取了特定音频片段，应用预定义条件（音量阈值 $\geq 33\text{dB}$ ，长度阈值 $\geq 2\text{s}$ ）。这些WAV文件的命名方式包含开始时间戳，增强了组织清晰度。随后，WAV文件数据集通过K-means算法进行了聚类处理。该算法应用于音频咳嗽数据，促进了咳嗽事件相关聚类的初步识别。为了优化和分类这些聚类，采用了一个10阶巴特沃斯滤波器，其截止频率范围为100 Hz至16kHz。由此产生的单维滤波声音数据被转换为二维梅尔频率倒谱系数（MFCC）数据，增强了声学特征的表示。进一步通过应用K-means算法进行优化，选择性地隔离与咳嗽声音相关的聚类。一项合作工作涉及

主题专家，住院医师

veterinarians, and aided by visible video cues, led to the manual labeling of each audio file based on the clustering results. The manual annotation encompassed diverse labels such as coughs, squeals, grunts, feeding sounds, human voices, door movements, chain sounds, and other ambient noise types. In the final steps of data preprocessing, video clips were systematically extracted from corresponding infrared video files, guided by the timestamps of the manually labeled sound clips. The culmination of these meticulous processes yielded a cohesive dataset of paired cough sound (WAV) and cough infrared (MP4) files. The standardized naming format for these files incorporated essential details, including the original audio/video file name, audio start timestamp, audio duration time, and a sequence number. The data distribution is shown in Fig 3.

Methods

Methodological overview

Coughing behavior, by its apparent features, produces an acoustic signal. From a physiological point of view, it will be accompanied by a change in body temperature, which will produce physiological signals. Therefore, this paper considers the influence of these two factors in recognizing cough. At the same time, the acoustic signal reflecting the biological characteristics and the deep features reflecting the physiological characteristics of body temperature are collected, and the alignment is integrated early to improve cough recognition performance. This study introduces a novel approach that utilizes significantly differentiated heterogeneous features to effectively perform automatic porcine cough identification. The flowchart of the proposed method is illustrated in Fig 4. Firstly, acoustic source biometric features are extracted from the preprocessed audio clips essential characteristics of the cough sounds. Secondly, thermal source deep physiological features, which reflect the distribution of body temperature, are extracted from thermal images using a lightweight shallow CNN. Thirdly, the extracted acoustic and physiological depth features are aligned and concatenated through early fusion to provide distinguishable multi-source heterogeneous fusion features. Finally, an SVM classifier is exploited to complete the classification of coughs and non-coughs.

Acoustic source biometric features

Using the audio processing method of the Librosa library, each sample of the cough audio clip in our constructed dataset was used as input, from which 29 acoustic features were extracted, and then recursive feature elimination combined with a cross-validation algorithm was used for feature selection, to select the acoustic source biometric feature to represent the cough sound signal. Time-domain analysis captures features in terms of time, frequency-domain analysis reveals features in terms of frequency, and cepstrum-domain analysis provides information on acoustic properties. Combining these results enables the extraction and classification of cough sounds, facilitating automated detection and recognition. In the time-domain analysis, the time-domain statistical features such as mean, variance, and energy are calculated to understand the temporal characteristics of cough sounds. Frequency domain analysis provides the energy distribution of the cough sound over different frequency components, thus revealing the frequency characteristics like frequency components and frequency range. Cepstrum domain analysis is a special frequency domain analysis method for extracting the acoustic characteristics from signals. Cepstrum domain analysis can provide the resonance characteristics of cough sounds, which helps to distinguish different types of cough sounds. By combining the analyses of the time domain, frequency domain, and cepstrum domain features, the characteristics of cough sounds can be revealed in a comprehensive way. The combinations of 29 acoustic source features consisting of the time, frequency, and cepstrum domains were

兽医，并借助可见视频提示，根据聚类结果对每个音频文件进行手动标注。手动标注涵盖了多种标签，如咳嗽声、吱吱声、咕噜声、喂食声、人声、门运动声、链条声以及其他环境噪音类型。在数据预处理最后步骤中，根据手动标注的声音片段的时间戳，从相应的红外视频文件中系统地提取视频片段。这些细致流程的最终成果是形成了一个包含配对咳嗽声（WAV）和咳嗽红外（MP4）文件的连贯数据集。这些文件的标准命名格式包含了关键信息，包括原始音频/视频文件名、音频起始时间戳、音频持续时间以及序列号。数据分布情况显示在 图3。

方法
方法学概述

咳嗽行为，凭借其明显特征，会产生声学信号。从生理学角度而言，它将伴随着体温变化，从而产生生理信号。因此，本文考虑了这两种因素在咳嗽识别中的影响。同时，收集反映生物特征的声学信号和反映体温生理特征的深度特征，进行早期对齐融合，以提升咳嗽识别性能。本研究介绍了一种利用显著差异的异构融合特征来有效执行自动猪咳嗽识别的新方法。所提出的方法流程图如图4所示。首先，从预处理后的音频片段中提取咳嗽声音的关键声源生物特征。其次，使用轻量级浅层CNN从热图像中提取反映体温分布的热源深度生理特征。第三，通过早期融合对齐并连接提取的声学 and 生理深度特征，提供可区分的多源异构融合特征。最后，采用SVM分类器完成咳嗽与非咳嗽的分类。图4。

声源生物特征

我们构建数据集和处理音频样本被用作输入，从中提取了29个声学特征，然后使用递归特征消除结合交叉验证算法进行特征选择，以选择声源生物特征来表示咳嗽声音信号。时域分析捕获时间方面的特征，频域分析揭示频率方面的特征，而倒谱域分析提供关于声学特性的信息。结合这些结果能够提取和分类咳嗽声音，促进自动检测和识别。在时域分析中，计算均值、方差和能量等时域统计特征，以了解咳嗽声音的时域特性。频域分析提供咳嗽声音在不同频率分量上的能量分布，从而揭示频率特征，如频率分量和频率范围。倒谱域分析是一种特殊的频域分析方法，用于从信号中提取声学特性。倒谱域分析可以提供咳嗽声音的共振特性，有助于区分不同类型的咳嗽声音。通过结合时域、频域和倒谱域特征的分析，可以全面揭示咳嗽声音的特性。由时域、频域和倒谱域组成的29个声源特征组合包括

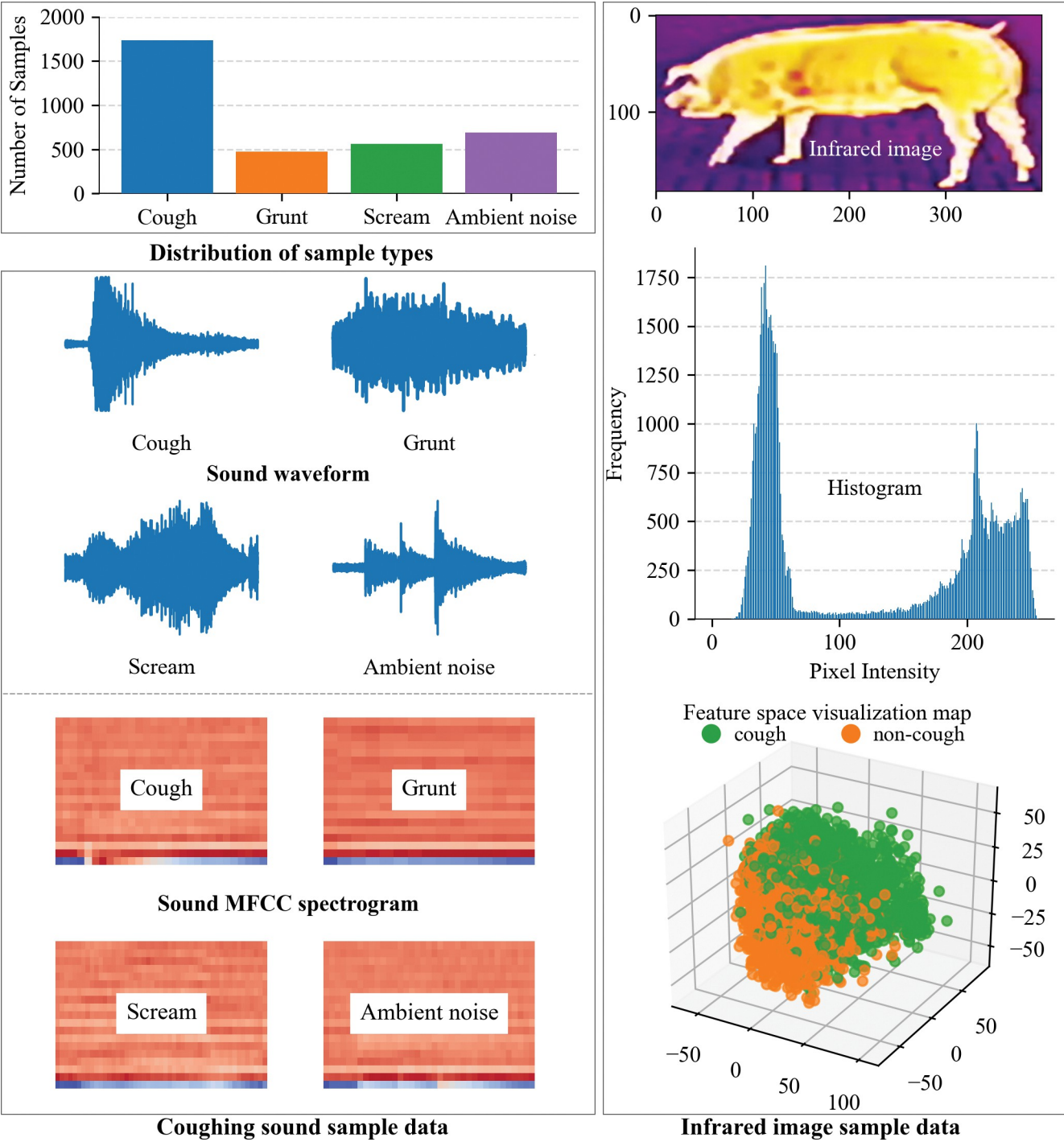


Fig 3. The data distribution graph.
<https://doi.org/10.1371/journal.pone.0297655.g003>

extracted from the one-dimensional sound signal. The time domain features include Root Mean Square (RMS) energy and Zero Crossing Rate (ZCR) [30]. The frequency domain features include spectral envelope, constant Q cepstral coefficients (CQCC), spectral centroid, spectral bandwidth, spectral rolloff, spectral flatness, and spectral flux [31]. Also, we extracted twenty MFCC cepstrum domains [32].

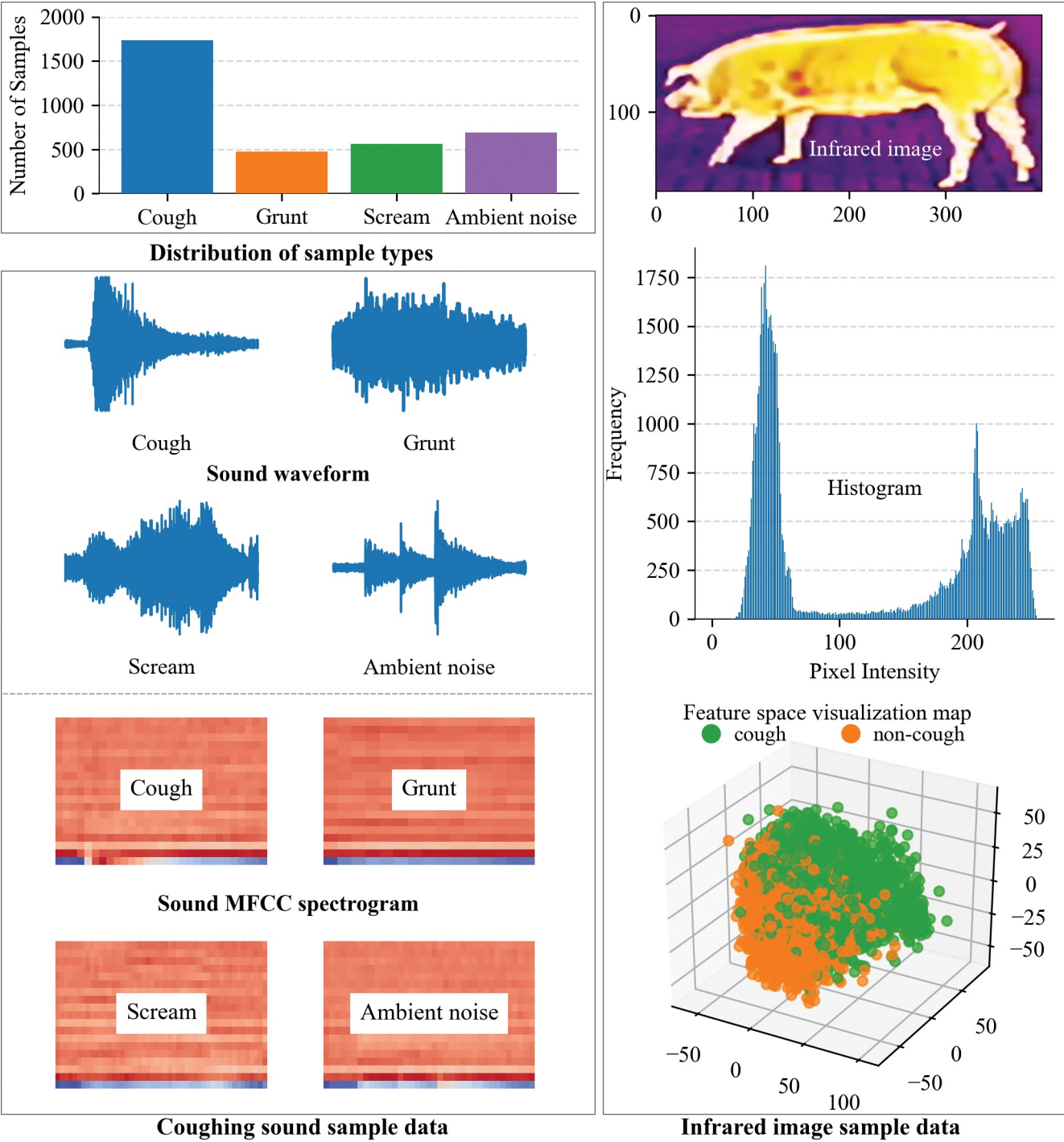


图3。数据分布图。
<<https://doi.org/10.1371/journal.pone.0297655.g003>>

从一维声音信号中提取的。时域特征包括均方根（RMS）能量和过零率（ZCR） [30]。频域特征包括频谱包络、恒Q倒谱系数（CQCC）、频谱质心、频谱带宽、频谱 rolloff、频谱平坦度和频谱通量 [31]。此外，我们提取了二十个MFCC倒谱域 [32]。

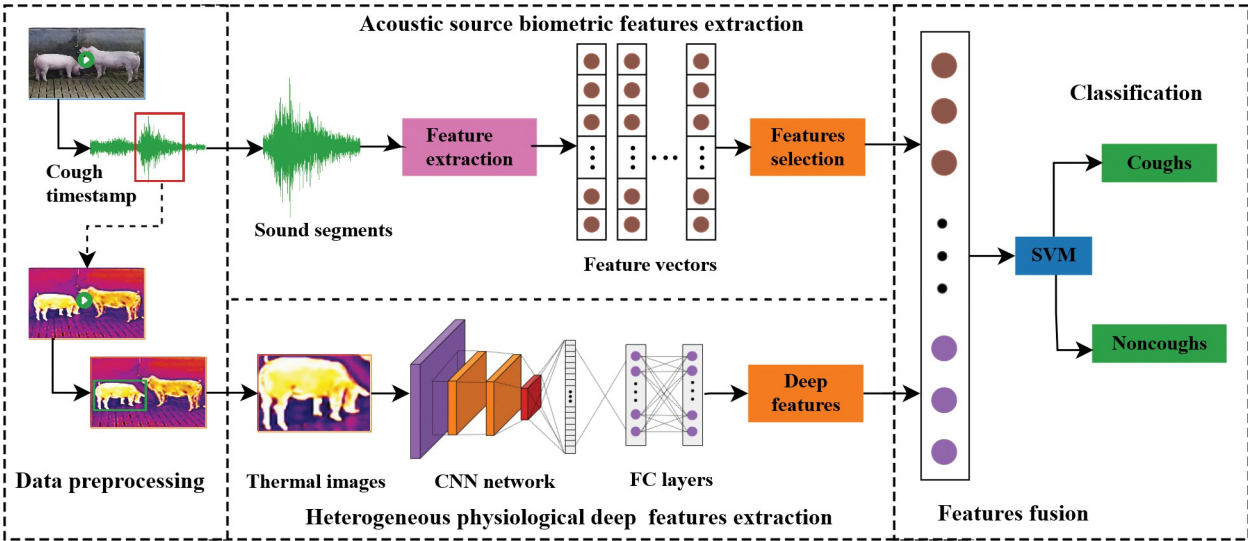


Fig 4. The flowchart of the proposed porcine cough recognition.

<https://doi.org/10.1371/journal.pone.0297655.g004>

Feature selection assumes paramount importance, especially within the realm of high-dimensional datasets like those characterizing coughing speech data. Prudent feature curation bolsters model interpretability, curbs overfitting tendencies, enhances computational efficiency, and ultimately elevates the precision of coughing event classification.

Support Vector Machine Recursive Feature Elimination Cross-Validation (SVM-RFECV) stands as a cutting-edge feature selection technique specifically tailored for the classification of coughing events utilizing speech data. This novel methodology synergizes the capabilities of Support Vector Machines (SVM) and Recursive Feature Elimination Cross-Validation (RFECV), as outlined by [33]. By iteratively eliminating noise data unrelated and redundant features through feature selection, we aim to improve the SVM model’s prediction accuracy and generalization ability [34].

Within the domain of coughing speech data, the potency of SVM-RFECV derives from the incorporation of SVM as the foundational classifier. SVMs are renowned for adeptly handling high-dimensional data and adeptly capturing intricate patterns. By entwining SVM with RFECV—a recursive feature elimination mechanism—SVM-RFECV systematically sifts and preserves the most pertinent attributes for classifying coughing events, eliminating irrelevant noise data (such as sustained high-frequency audio signals).

The ingenuity of SVM-RFECV hinges on its iterative feature pruning procedure, meticulously devised for the nuances of coughing speech data. The RFECV algorithm assesses feature importance through weight assignment during each iteration. Weight calculations, as delineated by Formula (1), coupled with the computation of importance levels as per Formula (2), facilitate the ranking of features. Subsequently, features with lower weights, indicative of diminished relevance, are progressively excised from the feature pool. The SVM model is then retrained using the winnowed feature ensemble. This iterative sequence perseveres until an optimum assemblage of features, germane to the classification of coughing events, is achieved [35].

$$w_i = |\alpha_i \times y_i|, i \in [1, P], \tag{1}$$

$$I(f_i) = n - k_i \tag{2}$$

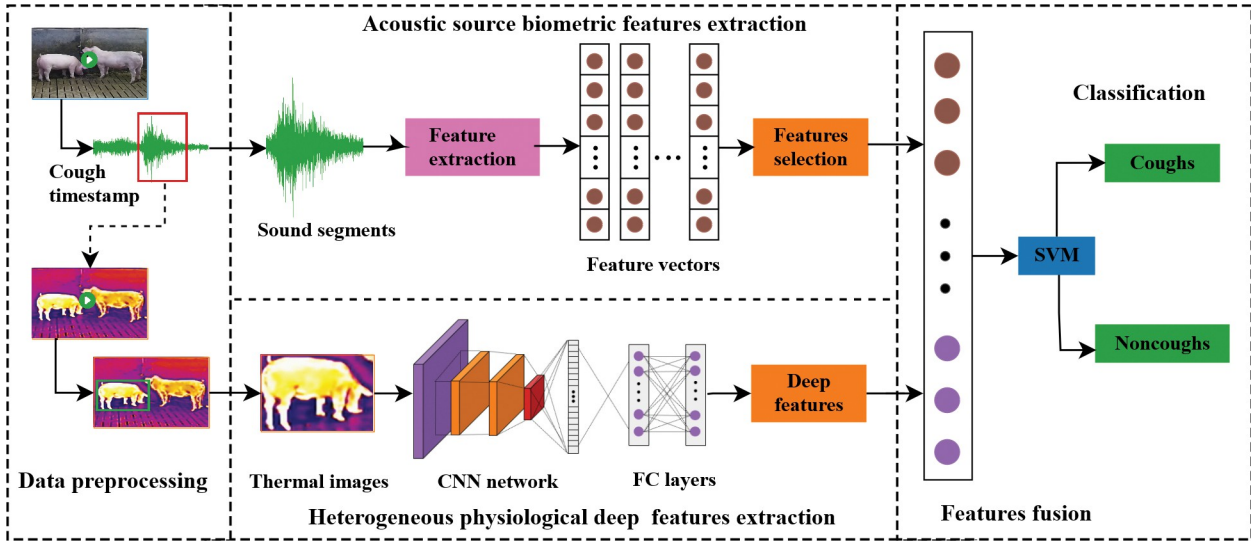


图 4. 所提出的猪咳嗽识别流程图。

<https://doi.org/10.1371/journal.pone.0297655.g004>

特征选择至关重要，尤其是在处理咳嗽语音数据这类高维数据集时。谨慎的特征筛选能够增强模型可解释性，抑制过拟合倾向，提高计算效率，并最终提升咳嗽事件分类的精确率。

支持向量机递归特征消除交叉验证（SVM-RFECV）是一种尖端特征选择技术，专门用于利用语音数据对咳嗽事件进行分类。这种新颖的方法结合了支持向量机（SVM）和递归特征消除交叉验证（RFECV）的功能，具体如下 [33]。通过迭代消除与冗余无关的噪声数据，我们旨在提高SVM模型的预测准确率和泛化能力 [34]。

在咳嗽语音数据的领域内，SVM-RFECV的强大之处源于将SVM作为基础分类器。SVM以其擅长处理高维数据和捕捉复杂模式而闻名。通过将SVM与RFECV（一种递归特征消除机制）结合，SVM-RFECV系统地筛选并保留对咳嗽事件分类最相关的属性，消除无关的噪声数据（例如持续的音频高频信号）。

SVM-RFECV 的巧妙之处在于其迭代特征剪枝过程，该过程为咳嗽语音数据的特性精心设计。RFECV 算法通过在每次迭代中分配权重来评估特征重要性。权重计算如公式（1）所述，结合根据公式（2）计算的重要性水平，促进特征的排序。随后，权重较低、表明相关性减弱的特征会逐步从特征集中移除。然后使用筛选后的特征集重新训练 SVM 模型。这一迭代过程持续进行，直到达到一个与咳嗽事件分类相关的最佳特征组合[35]。

$$w_i = |\alpha_i \times y_i|, i \in [1, P], \tag{1}$$

$$I(f_i) = n - k_i \tag{2}$$

where f_i denotes the i th feature in the dataset, w_i denotes the weight of the i th feature, P denotes the number of features (where P is set to 29), y_i denotes the classification class of the i th feature, $I(f_i)$ denotes the importance of the i th feature, n denotes the number of samples, and k_i denotes the number of times the i th features are removed in the recursive feature elimination process. This process removes redundant acoustic features through sound source feature selection. The highest predicted F1 score achieves the optimal feature combination to generate a new set of acoustic source biometric features.

Thermal source deep physiological features

The occurrence of coughing behavior in porcine respiratory infections triggers discernible alterations in the infrared body temperature signal of the pigs. Particularly noteworthy is the distinctive variance in body temperature between coughing pigs and their healthy counterparts. Coughing pigs manifest elevated body temperatures that exhibit unique infrared signal patterns. This observation underpins the efficacy of a method involving the acquisition of infrared images from coughing pigs, extraction of infrared deep features using a Convolutional Neural Network (CNN), and feeding these features into an SVM classification network. The outcome is a non-contact, automated system capable of detecting pig coughing events.

Physiological deep features, or high-level features beyond epistemic knowledge, are derived through deep learning techniques applied to raw thermal images of physiological signals. These features hold immense potential across various applications, including diagnostics and disease prediction [36]. In our study, we harnessed a lightweight shallow CNN architecture to extract infrared deep physiological features that faithfully mirror real-time surface temperatures of monitored pigs.

The integration of infrared thermal images within our approach is motivated by two pivotal factors. Firstly, infrared thermal images aptly capture the distribution of body surface temperatures while encapsulating the physiological nuances of living organisms [37]. Their efficacy in automated porcine cough event identification has been well-documented [38]. Secondly, the shallow CNN architecture proves well-matched for feature extraction from thermal images [39]. The amalgamation of this architecture with SVM classification engenders highly accurate classification outcomes while significantly boosting execution speed [40].

Our methodology encompasses the gathering of infrared images from coughing pigs, the extraction of infrared depth features via a CNN network, and their subsequent input into an SVM classification network. This non-contact approach translates into an automated mechanism for detecting pig coughing, thereby delivering an exhaustive and robust analysis of the physiological attributes associated with coughing events.

Different shallow convolutional neural networks have their own strengths. LeNet-5 is a classical convolutional neural network model proposed by Yann LeCun et al. in 1998, which may not effectively capture higher-level, more abstract feature information for complex image classification tasks [41]. AlexNet was proposed by Alex Krizhevsky et al. in 2012 and uses the ReLU activation function to nicely mitigate the gradient disappearance problem and performs well in large-scale image classification tasks but is prone to overfitting and requires significant time and resources for training and inference [42]. Compared to other deeper architectures such as VGG [43] and ResNet [44], where LeNet-5 is a shallow network more suitable for smaller datasets, AlexNet is a deeper network for extracting higher-level feature information. Our experiments are on a private and limited dataset where higher-level deep features must be extracted for evaluation. The deployment requires an optimal solution between recognition speed, model size, and performance.

在 f_i 表示数据集中第 i 个特征, w_i 表示第 i 个特征的权重, P 表示特征数量 (其中 P 设置为 29), y_i 表示第 i 个特征的分类类别, $I(f_i)$ 表示第 i 个特征的重要性, n 表示样本数量, 以及 k_i 表示在递归特征消除过程中第 i 个特征被移除的次数。该过程通过声源特征选择来移除冗余的声学特征。最高的预测 F1 分数实现了最优特征组合, 以生成一套新的声源生物特征。

热源深度生理特征

猪呼吸道感染中咳嗽行为的出现会引发猪红外体温信号的明显变化。尤其值得注意的是, 咳嗽猪与健康猪之间体温存在显著差异。咳嗽猪表现出体温升高, 并呈现独特的红外信号模式。这一观察结果证实了以下方法的有效性: 从咳嗽猪获取红外图像, 使用卷积神经网络 (CNN) 提取红外深度特征, 并将这些特征输入支持向量机 (SVM) 分类网络。最终形成一种非接触式自动化系统, 能够检测猪咳嗽事件。

生理深度特征, 即超越认知知识的高级特征, 是通过深度学习技术应用于生理信号原始红外图像而提取的。这些特征在诊断和疾病预测等多种应用中具有巨大潜力 [36]。在我们的研究中, 我们采用轻量级浅层 CNN 架构提取红外深度生理特征, 这些特征忠实地反映了被监测猪的实时表面温度。

我们的方法中整合红外热成像是基于两个关键因素。首先, 红外热成像能够恰当地捕捉生物体表面温度的分布, 同时体现其生理特征 [37]。其在自动识别猪咳嗽事件方面的有效性已被充分证实 [38]。其次, 浅层卷积神经网络架构非常适合从热成像中提取特征[39]。这种架构与支持向量机分类的融合产生了高度精确的分类结果, 同时显著提升了执行速度 [37]。

— 这种架构与支持向量机分类的融合产生了高度精确的分类结果, 同时显著提升了执行速度 [40]。

我们的方法论包括从咳嗽的猪身上收集红外图像, 通过卷积神经网络提取红外深度特征, 并将这些特征输入到支持向量机分类网络中。这种非接触式方法转化为一种自动检测猪咳嗽的机制, 从而对与咳嗽事件相关的生理属性进行详尽且稳健的分析。

不同的浅层卷积神经网络各有其优势。LeNet-5 是 Yann LeCun 等人在 1998 年提出的经典卷积神经网络模型, 但在复杂图像分类任务中可能无法有效捕捉高级、更抽象的特征信息 [41]。AlexNet 是 Alex Krizhevsky 等人在 2012 年提出的, 它使用 ReLU 激活函数很好地缓解了梯度消失问题, 在大规模图像分类任务中表现良好, 但容易过拟合, 并且训练和推理需要大量时间和资源 [42]。与其他更深的架构 (如 VGG [43] 和 ResNet [44],) 相比, LeNet-5 是一个浅层网络, 更适合小数据集, 而 AlexNet 是一个更深层的网络, 用于提取高级特征信息。我们的实验基于一个私有且有限的数据集, 需要提取高级深度特征进行评估。部署需要在识别速度、模型大小和性能之间找到最佳解决方案。

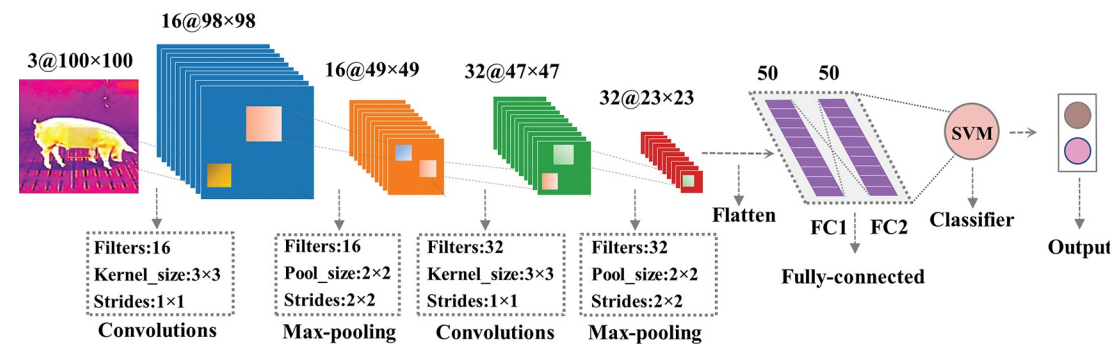


Fig 5. ThermographicNet for extracting thermal source deep physiological features.

<https://doi.org/10.1371/journal.pone.0297655.g005>

Therefore, we propose a lightweight shallow convolutional neural network architecture, named ThermographicNet, as a deep physiological feature extractor. ThermographicNet combines the advantageous design elements of the Lenet-5 and Alexnet architecture, as illustrated in Fig 5.

The proposed ThermographicNet architecture incorporates specific design choices to effectively process thermal images for porcine cough recognition. The network comprises several key layers and operations, as described below:

1. Input layer: The network takes a 100×100 pixel thermal image as input.
2. Convolutional layers: The first convolutional layer consists of 16 filters with a size of 3×3, and the second convolutional layer includes 32 filters of the same size. Both convolutional layers apply Rectified Linear Unit (ReLU) activation functions, which enables faster convergence and avoids gradient vanishing.
3. Max-pooling layers: Two max-pooling layers with a size of 2×2 follow the convolutional layers. These layers downsample the spatial dimensions of the feature maps, capturing the most salient information while reducing computational complexity.
4. Flattening layer: The output of the previous layer is flattened into a one-dimensional vector.
5. Dropout layer: It is designed to randomly deactivate a fraction of neurons with a probability of 0.5 to alleviate the overfitting problem by enhancing the network’s generalization capability.
6. Fully Connected (FC) layers: Two FC layers with 50 neurons each are connected to the Dropout layer.

2D convolution was chosen to convolve the different thermal physiological features simultaneously as the thermal distribution of the thermal image is a 3-channel 2D image. ReLU was chosen because it converges faster and does not suffer from gradient disappearance. The Dropout layer was chosen to increase the generalization capability of the network in order to prevent possible overfitting problems for thermal images extracted within a small time slice (≈ 2.13 s) of a persistent cough. Adam was chosen as the optimizer for model training, Sparse_Categorical_Crossentropy was chosen as the loss function, and accuracy was used as the model evaluation metric.

The performance of two fully connected layers of Lenet-5, AlexNet, DenseNet121, Vgg16, Vgg19, ResNet50, ResNet101, ResNet152, and our proposed ThermographicNet network are first evaluated separately for deep physiological feature extraction. Then, the two fully

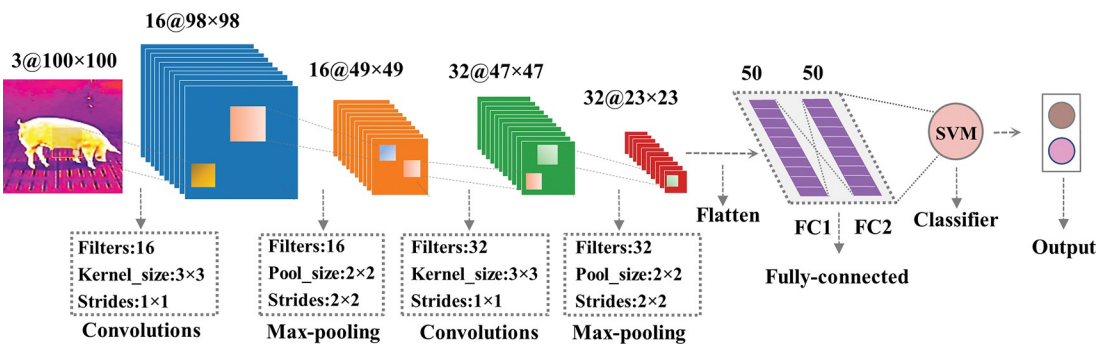


图 5. 热成像网络用于提取热源深度生理特征。

<https://doi.org/10.1371/journal.pone.0297655.g005>

因此，我们提出了一种轻量级的浅层卷积神经网络架构，命名为 ThermographicNet，作为深度生理特征提取器。ThermographicNet 结合了 LeNet-5 和 AlexNet 架构的有利设计元素，如图 5 所示Fig 5。

所提出的ThermographicNet架构通过特定的设计选择，有效地处理热成像图像以实现猪咳嗽识别。该网络包含多个关键层和操作，具体描述如下：

1. 输入层：网络以一个 100×100像素的热成像图像作为输入。
2. 卷积层：第一个卷积层包含16个大小为 3×3的滤波器，第二个卷积层包含32个相同大小的滤波器。这两个卷积层都应用了修正线性单元（ReLU）激活函数，这可以实现更快的收敛并避免梯度消失。
3. 最大池化层：两个大小为 2×2的最大池化层紧随卷积层之后。这些层对特征图的空间维度进行下采样，在捕获最显著信息的同时减少计算复杂度。
4. 扁平化层：前一层的输出被展平为一个一维向量
5. Dropout层：它被设计为以0.5的概率随机禁用一部分神经元，通过增强网络泛化能力来缓解过拟合问题。
6. 全连接（FC）层：两个各包含50个神经元的全连接层与Dropout层相连。

2D卷积被选择用于同时卷积不同的热生理特征，因为热图像的热分布是一个3通道的2D图像。ReLU被选择是因为它收敛更快，并且不会遭受梯度消失。Dropout层被选择用于提高网络的泛化能力，以防止在持续咳嗽（时间片内为2.13秒）中提取的热图像可能出现的过拟合问题。Adam被选择作为模型训练的优化器，Sparse_Categorical_Crossentropy被选择作为损失函数，准确率被用作模型评估指标。

首先分别评估了Lenet-5、AlexNet、DenseNet121、Vgg16、Vgg19、ResNet50、ResNet101、ResNet152以及我们提出的ThermographicNet网络两个全连接层的性能，用于深度生理特征提取。

全连接层是分别用于深度生理特征提取。然后，两个全

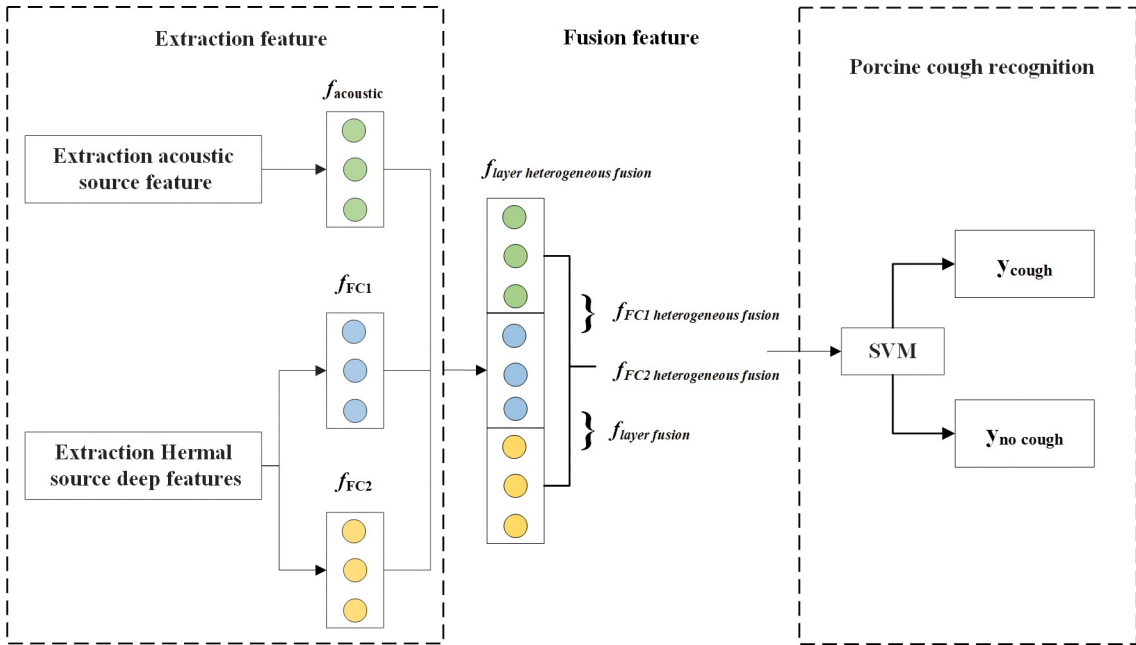


Fig 6. Flowchart of early feature fusion technique with four different fusion strategies.

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connected layer feature vectors extracted from the ThermographicNet network were aligned and concatenated to form the layer fusion deep physiological features for classification evaluation. Finally, the FC1 feature vector extracted from the ThermographicNet network was selected as the thermal source of deep physiological features.

Feature fusion methods

Extracting acoustic biometric features and deep physiological features from audio and infrared data, respectively, and fusing them to provide distinguishable multimodal fusion features for accurate automatic cough recognition to break through the performance bottleneck of cough sound classification. The fusion method adopts feature-level fusion, also known as early fusion, which is the most commonly used strategy in multimodal recognition systems. After extraction, it means immediately connecting features extracted from different modalities into a single high-dimensional feature vector. We have provided four different fusion strategies to evaluate the cough classification task. Specifically, first, obtain the acoustic source biometric features ($f_{acoustic}$) from the acoustic source, the deep physiological features (f_{FC1}) from the first fully connected layer of the ThermographicNet network, and the physiological deep feature (f_{FC2}) from the second fully connected layer. Then, concatenate f_{FC1} and f_{FC2} to obtain $f_{layer fusion}$, concatenate $f_{acoustic}$ and f_{FC1} to obtain $f_{FC1 heterogeneous fusion}$, concatenate $f_{acoustic}$ and f_{FC2} to obtain $f_{FC2 heterogeneous fusion}$, and concatenate $f_{acoustic}$ and $f_{layer fusion}$ to obtain $f_{layer heterogeneous fusion}$. Finally, input these features into an SVM classifier to obtain the classification results for the cough task, as shown in Fig 6.

Classification

In the porcine cough recognition classification task, selecting an appropriate classifier is crucial to handling the extracted features effectively. The early fusion technique is employed to extract highly representative nonlinear and high-dimensional feature vectors that capture the essential

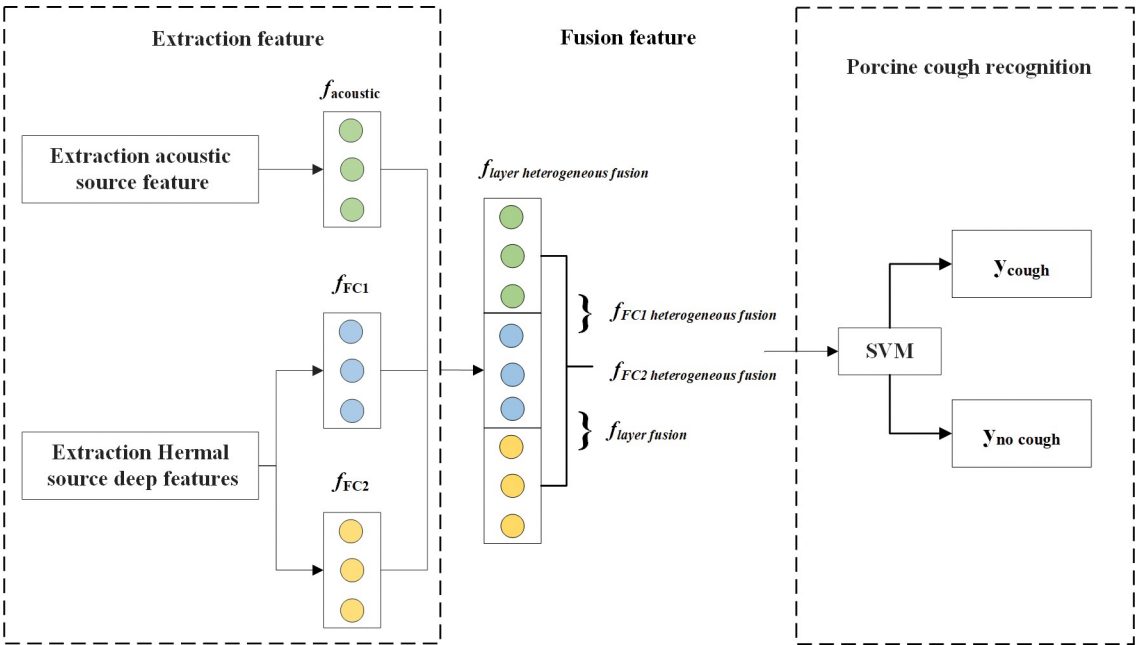


图6。早期特征融合技术流程图，包含四种不同的融合策略。

<https://doi.org/10.1371/journal.pone.0297655.g006>

从热成像网络中提取的连接层特征向量被对齐并连接，形成用于分类评估的层融合深度生理特征。最后，从热成像网络中提取的FC1特征向量被选为深度生理特征的热源。

特征融合方法

分别从音频和红外数据中提取声源生物特征和深度生理特征，并将它们融合以提供可区分的多模态融合特征，用于精确的自动咳嗽识别，突破咳嗽声分类的性能瓶颈。该融合方法采用特征级融合，也称为早期融合，这是多模态识别系统中最常用的策略。提取后，意味着立即将来自不同模态的特征连接到一个高维特征向量中。我们提供了四种不同的融合策略来评估咳嗽分类任务。具体来说，首先，从声源获取声源生物特征 ($f_{acoustic}$)，从ThermographicNet网络的第一层全连接层获取深度生理特征 (f_{FC1})，以及从第二层全连接层获取生理深度特征 (f_{FC2})。然后，连接 f_{FC1} 和 f_{FC2} 以获得 $f_{layer fusion}$ ，连接 $f_{acoustic}$ 和 f_{FC1} 以获得 $f_{FC1 异构融合}$ ，连接 $f_{acoustic}$ 和 f_{FC2} 以获得 $f_{FC2 异构融合}$ ，以及连接 $f_{acoustic}$ 和 $f_{layer fusion}$ 以获得 $f_{layer 异构融合}$ 。最后，将这些特征输入SVM分类器以获得咳嗽任务的分类结果，如图6所示。

分类

在猪咳嗽识别分类任务中，选择合适的分类器对于有效处理提取的特征至关重要。采用早期融合技术来提取具有高度代表性的非线性、高维特征向量，这些特征向量能够捕捉咳嗽从多种模态中提取的关键特征。

features of cough from multiple modalities. These feature vectors are then concatenated and used as input for the classifier. Considering the requirements of handling high-dimensional feature spaces and nonlinear decision boundaries, the SVM (Support Vector Machine) classifier emerges as an advantageous choice.

The SVM classifier possesses the necessary strengths to handle the complexities of the cough recognition task. It excels in effectively processing high-dimensional feature spaces and modeling nonlinear decision boundaries. Leveraging these capabilities, the SVM classifier accurately classifies cough events even in intricate and overlapping patterns. By utilizing the SVM classifier, we enhance the performance and reliability of the cough recognition system, contributing to the advancement of automatic cough identification technology.

Experiments and results

The experimental datasets are constructed in this Section, and each experiment’s model evaluation metrics, implementation method, main execution environments configuration, and results are given.

Datasets

We will create our dataset using the preprocessed pig cough audio files and thermal imaging video files obtained from Data acquisition and preprocessing Section. We load the corresponding thermal imaging video files based on the cough audio file names. We extract the thermal image video clip from each thermal imaging video file. A Python script is used to extract five frames of image data from each video clip, which serve as candidate thermal source images.

With the assistance of the visible video and the resident veterinarian, the expert selects one image from the candidate thermal source images as the fusion target image. The selected fusion target image is then labeled using the LabelMe software [45]. The extracted thermal image file is named based on the corresponding audio file and marked as a positive sample.

Similarly, the expert follows the same procedure for negative samples to select the target pig image as the thermal image corresponding to the non-cough audio. We extract 1736 cough sounds and their corresponding 1736 physiological thermal images, which form the positive sample set. The negative sample set also consists of 1731 non-cough audio fragments (477 grunts, 565 screams, and 689 ambient noises) and their corresponding 1731 thermal images.

Metrics

The dataset is divided into a 70% training set and a 30% test set with no duplicate samples between the two sets. To evaluate the performance of our proposed method, a ten-fold cross-validation method with grid search is utilized on the training set. It explores and determines the optimal set of hyperparameters for training the SVM algorithm. The initial values of the hyperparameters are set to set with default values, and the corresponding models are generated based on the preprocessed training set. The test set is then used to evaluate the performance of the best pre-trained model obtained from the grid search. This grid search procedure allows us to systematically explore different combinations of hyperparameters and select the ones that yield the best performance for our cough recognition task. In the grid search process, for linear models, C is set to 0.9 and 1. For nonlinear models, the radial basis function (RBF) is used as the kernel function, and the value of Gamma is computed by Formula (3) [46], with the value of C set to the same value as the linear models.

$$Gamma = \frac{1}{n_{features} \times X.var()}$$

(3)

然后，这些特征向量被连接起来并用作分类器的输入。考虑到处理高维特征空间和非线性决策边界的要求，SVM（支持向量机）分类器成为一个有利的选择。

支持向量机分类器具备处理咳嗽识别任务复杂性的必要优势。它在有效处理高维特征空间和建模非线性决策边界方面表现出色。利用这些能力，支持向量机分类器能够准确分类咳嗽事件，即使在错综复杂和重叠的模式中也能如此。通过使用支持向量机分类器，我们提升了咳嗽识别系统的性能和可靠性，为自动咳嗽识别技术的进步做出了贡献。

实验与结果

本节构建了实验数据集，并给出了每个实验的模型评估指标、实现方法、主要执行环境配置和结果。

数据集

我们将使用从数据采集与预处理节段获得的预处理猪咳嗽音频文件和热成像视频文件来创建数据集。我们根据咳嗽音频文件名加载相应的热成像视频文件。我们从每个热成像视频文件中提取热图像视频片段。使用Python脚本从每个视频片段中提取五帧图像数据，这些图像数据作为候选热源图像。

在可见视频和驻场兽医的协助下，专家从候选热源图像中选择一张图像作为融合目标图像。然后使用LabelMe软件对选定的融合目标图像进行标记 [45]。提取的热图像文件根据相应的音频文件命名，并标记为正样本。

类似地，专家对负样本遵循相同流程，选择目标猪图像作为对应非咳嗽音频的热图像。我们提取了1736个咳嗽声音及其对应的1736张生理热图像，形成正样本集。负样本集也包含1731个非咳嗽音频片段（477个咕噜声、565个尖叫和689个环境噪音）及其对应的热图像。

指标

数据集被划分为70%的训练集和30%的测试集，且两个集合之间无重复样本。为评估我们提出的方法的性能，在训练集上采用网格搜索的十折交叉验证方法。该方法探索并确定用于训练SVM算法的最佳超参数组合。超参数的初始值设置为默认值，并基于预处理后的训练集生成相应的模型。然后使用测试集评估网格搜索得到的最佳预训练模型的性能。网格搜索过程使我们能够系统地探索不同的超参数组合，并选择那些在我们咳嗽识别任务中表现最优的组合。在网格搜索过程中，对于线性模型，C设置为0.9和1。对于非线性模型，使用径向基函数（RBF）作为核函数，Gamma的值通过公式（3） [46], 计算得出，其中C的值与线性模型相同。

$$Gamma = \frac{1}{n_{features} \times X.var()}$$

(3)

where X represents the training sample, $n_features$ stands for the number of features in the input sample, and $var()$ means the variance of the training set samples.

After training the model, its classification performance is evaluated using the test set. The evaluation metrics used for our model include accuracy, recall, precision, and F1-score (see Formulas (4)–(7). Here, we define cough as a positive sample and non-cough as a negative sample. True positives (TP) refer to correctly classified cough samples, true negatives (TN) represent correctly classified non-cough samples, false positives (FP) indicate non-cough samples misclassified as cough, and false negatives (FN) represent cough samples misclassified as non-cough. The evaluation metrics are calculated as follows:

$$Accuracy = \frac{TP + TN}{(TP + TN + FP + FN)} \tag{4}$$

$$Precision = \frac{TP}{(TP + FP)} \tag{5}$$

$$Recall = \frac{TP}{(TP + FN)} \tag{6}$$

$$F_1 - Score = \frac{2 \times Precision \times Recall}{(Precision + Recall)} \tag{7}$$

Implementation

The primary execution environments and parameters configured for our experiments are listed in Table 1. The program is implemented based on the Python language, the LibROSA library, and the Keras deep learning framework.

Table 1. Description of main execution environments and parameter configurations.

Execution Environment	Parameter
CPU	Intel Xeon Gold 6139M With 2.30GHz
Memory	251GB
GPU	NVIDIA Corporation GA102 [GeForce RTX 3090] (rev a1)
GPU Memory	24GB
Cuda	11.2
Anaconda	23.1.0
Python	3.9.7
LibROSA	0.9.2
Scikit-learn	1.1.3
Keras Version	2.5.0
Tensorflow-GPU	2.5.0
Optimizer	Adam optimizer
Batch Size	32
Learning_Rate	0.001
MFCC_Dim	20
Dropout_Prob	0.5
Epoch_Max	50
Storage Device	Samsung SSD 870 With 4TB
Operating System	Linux Ubuntu SMP 18.04.1

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其中X表示训练样本， $n_特征$ 表示输入样本中的特征数量， $var()$ 表示训练集样本的方差。

在训练模型后，使用测试集评估其分类性能。我们模型的评估指标包括准确率、召回率、精确率和F1分数（参见公式（4）–（7）。在此，我们将咳嗽定义为阳性样本，非咳嗽定义为阴性样本。真正例（TP）指正确分类的咳嗽样本，真负例（TN）指正确分类的非咳嗽样本，假正例（FP）指被误分类为咳嗽的非咳嗽样本，假负例（FN）指被误分类为非咳嗽的咳嗽样本。评估指标的计算方法如下：

$$Accuracy = \frac{TP + TN}{(TP + TN + FP + FN)} \tag{4}$$

$$Precision = \frac{TP}{(TP + FP)} \tag{5}$$

$$Recall = \frac{TP}{(TP + FN)} \tag{6}$$

$$F_1 - Score = \frac{2 \times Precision \times Recall}{(Precision + Recall)} \tag{7}$$

实施

我们实验配置的主要执行环境和参数列于表1。该程序基于Python语言、LibROSA库和Keras深度学习框架实现。

表1. 主要执行环境和参数配置的描述。

执行环境	参数
CPU	英特尔至强金牌6139M 2.30GHz
内存	251GB
GPU	英伟达公司 GA102 [GeForce RTX 3090] (rev a1)
显存	24GB
Cuda	11.2
Anaconda	23.1.0
Python	3.9.7
LibROSA	0.9.2
Scikit-learn	1.1.3
Keras 版本	2.5.0
Tensorflow-GPU	2.5.0
优化器	Adam优化器
批大小	32
学习_率	0.001
MFCC_Dim	20
Dropout_概率	0.5
Epoch_最大值	50
存储设备	三星 SSD 870 配备 4TB
操作系统	Linux Ubuntu SMP 18.04.1

<https://doi.org/10.1371/journal.pone.0297655.t001>

The automatic framework for pig cough recognition is implemented using the Python programming language. The processing and feature extraction of cough audio data is implemented using the Librosa library. The processing and physiological deep feature extraction of cough infrared image data are implemented using the Keras deep learning library, including the design, debugging, evaluation, application, and visualization of CNN network models. Fusion and classification are implemented using the scikit-learn machine learning toolkit. Grad-CAM (Gradient Weighted Class Activation Mapping) [47] is used to interpret the ThermographicNet model visually.

The cough audio data processing program mainly includes preprocessing, feature extraction, and feature selection of cough data. Preprocessing and feature extraction primarily utilize libraries such as Librosa, NumPy, CSV, and OS. The cough audio has a sampling rate of 48000Hz, followed by the extraction of 29 sound features using functions such as zero_crossing_rate, RMS, chroma_stft, chroma_cqt, spectral_centroid, spectral_bandwidth, spectral_contrast, spectral_rolloff, spectral_flatness, onset_strength, and MFCC provided by the Librosa feature class. In the feature selection, libraries such as scikit-learn, pandas, and matplotlib are mainly used. The data set is split into training and testing sets using the train_test_split method in the model_selection class of the scikit-learn package, with the testing set accounting for 30%. Recursive feature elimination and 5-fold and 10-fold cross-validation are performed using the RFECV function in the model_selection class. Finally, the cough sounds are classified using the SVC function in the SVM class of the scikit-learn library.

Processing cough infrared image data mainly involves network design, debugging, and evaluation. Using libraries such as Keras, scikit-learn, pandas, NumPy, and OS, the infrared image data is resized to 100×100 pixels. The data set is split into training and testing sets using the train_test_split method in the model_selection class of the scikit-learn package, with the testing set accounting for 30%. Subsequently, network models, including Lenet-5, AlexNet, DenseNet121, Vgg16, Vgg19, ResNet50, ResNet101, ResNet152, and ThermographicNet, are designed using the Keras library and trained and evaluated. The optimizer used is Adam, with a maximum number of epochs (epoch_max) set to 100, a batch size of 32, a learning rate of 0.001, and 20% of the training set used as a validation set for network evaluation. After training all the network models, the load_model function in the models class of the Keras library is used to load the models. The deep features of the infrared images are predicted using the models, and the FC layer physiological deep features data are extracted for subsequent deep feature fusion.

The scikit-learn, Keras, pandas, and NumPy libraries are primarily used to perform three fusion experiments in the feature fusion and classification experiments. The fusion function is implemented using the concat function provided by the pandas library. Classification is performed using the SVC function in the SVM class of the scikit-learn library. The SVM kernel is set to rbf (radial basis function), with a penalty parameter C of 1.0 and Gamma set to ‘auto’. After training the SVM model, the cross_val_score function in the model_selection class of the scikit-learn library is used to perform 10-fold cross-validation to obtain performance evaluation metrics of the classifier, including accuracy, precision, recall, and F1-score. Throughout all the data processing procedures mentioned above, the StandardScaler function in the preprocessing class of the scikit-learn library is used to normalize the feature data.

Results

This section primarily encompasses four experiments. The Evaluation of acoustic source biometric features aims to assess the extraction of optimal acoustic source features and evaluate the cough sound classification performance using single acoustic features by inputting them

猪咳嗽识别的自动框架使用Python编程语言实现。咳嗽音频数据的处理和特征提取使用Librosa库实现。咳嗽红外图像数据的处理和生理深度特征提取使用Keras深度学习库实现，包括卷积神经网络模型的设计、调试、评估、应用和可视化。融合和分类使用scikit-learn机器学习工具包实现。Grad-CAM（梯度加权类激活映射）[47]用于可视化ThermographicNet模型。

咳嗽音频数据处理程序主要包含咳嗽数据的预处理、特征提取和特征选择。预处理和特征提取主要利用Librosa、NumPy、CSV和OS等库。咳嗽音频的采样率为48000Hz，随后使用Librosa特征类提供的零交叉率_、RMS、chroma_stft、chroma_cqt、频谱_质心、频谱_带宽、频谱_对比度、频谱_rolloff、频谱_平坦度、起音_强度和MFCC等函数提取29个声音特征。在特征选择中，主要使用scikit-learn、pandas和matplotlib等库。利用scikit-learn包的模型_选择类中的train_test_split方法将数据集分为训练集和测试集，测试集占比30%。使用模型_选择类中的RFECV函数进行递归特征消除和5折及10折交叉验证。最后，使用scikit-learn库的SVM类中的SVC函数对咳嗽声音进行分类。

处理咳嗽红外图像数据主要涉及网络设计、调试和评估。使用 Keras、scikit-learn、pandas、NumPy 和 OS 等库，将红外图像数据调整大小为 100×100 像素。使用 scikit-learn 包中模型选择类中的 train_test_split 方法将数据集分为训练集和测试集，测试集占比 30%。随后，使用 Keras 库设计包括 LeNet-5、AlexNet、DenseNet121、Vgg16、Vgg19、ResNet50、ResNet101、ResNet152 和热成像网络等网络模型，并进行训练和评估。优化器

使用的是 Adam，最大 epoch 数量（epoch_max）设置为 100，批大小为 32，学习率为 0.001，20% 的训练集用作网络评估的验证集。训练所有网络模型后，使用 Keras 库中 models 类中的 load_model 函数加载模型。使用模型预测红外图像的深度特征，并提取 FC 层生理深度特征数据，用于后续的深度特征融合。

scikit-learn、Keras、pandas 和 NumPy 库主要用于特征融合和分类实验中的三项融合实验。融合功能使用 pandas 库提供的 concat 函数实现。分类使用 scikit-learn 库的 SVM 类中的 SVC 函数执行。SVM 核设置为 rbf（径向基函数），惩罚参数 C 为 1.0，Gamma 设置为 ‘auto’。在训练 SVM 模型后，使用 scikit-learn 库的 model_selection 类中的 cross_val_score 函数进行 10 折交叉验证，以获得分类器的性能评估指标，包括准确率、精确率、召回率和 F1 分数。在上述所有数据处理过程中，使用 scikit-learn 库的 preprocessing 类中的 StandardScaler 函数对特征数据进行归一化处理。

结果

本节主要涵盖四项实验。声源生物特征评估旨在输入数据以评估最佳声源特征提取，并使用单一声学特征输入支持向量机分类器来评估咳嗽声音分类性能。

into an SVM classifier. The Evaluation of thermal source deep physiological features involves selecting the optimal feature extractor for extracting deep physiological features from thermal sources and evaluating the cough classification capability using single deep physiological features from infrared data by inputting them into an SVM classifier. The Evaluation of feature fusion combines the optimal features extracted from the previous two experiments using different fusion strategies. It evaluates the cough classification performance of the fused features by inputting them into an SVM classifier. Finally, the Evaluation of recognition speed and model size compares and evaluates the selected optimal methods’ recognition speed and model size from the previous three experiments.

Evaluation of acoustic source biometric features

This experiment aims to evaluate the performance of the SVM-RFECV algorithm and select the optimal feature set as acoustic source biometric features from a pool of 29 cough sound features based on the experimental results. These selected features will then be inputted into an SVM classifier to assess the classification performance of single acoustic features for cough recognition. When comparing SVM-RFECV with other feature selection methods, we considered several factors that led to our choice of SVM-RFECV. SVM-RFECV offers an iterative process that eliminates irrelevant or redundant features, effectively reducing the impact of noise and enhancing the model’s robustness. Additionally, SVM-RFECV leverages the power of Support Vector Machines (SVMs), known for their ability to handle high-dimensional data and capture complex patterns. We achieve accurate feature selection by combining SVM with RFECV while maintaining computational efficiency. These advantages make SVM-RFECV a suitable choice for our specific application and contribute to its selection over alternative feature selection methods.

Regarding feature selection metrics, we employed the F1 score, which considers both precision and recall, providing a comprehensive assessment of the model’s performance. By evaluating the F1 score, we can effectively measure the trade-off between precision and recall, ensuring the selected features optimize the classification results.

When employing SVM-RFECV for feature selection, we evaluated the number of features in the model and their respective performance using cross-validation test scores. Specifically, Figs 7 and 8 present the evaluation results using five-fold and ten-fold cross-validation, respectively. In Fig 7, we observe that an improved F1 score is achieved when utilizing eight features, and the F1 scores for both training and validation data tend to stabilize. Subsequently, the model’s accuracy gradually improves as the number of features increases. The optimal model evaluation metrics (Accuracy 92.13%, Precision 92.18%, Recall 92.05%, F1-score 92.10%) were reached when the number of features increased to 22. In Fig 8, the overall trend is consistent with Fig 7, and the optimal model evaluation metrics (Accuracy 91.84%, Precision 91.83%, Recall 91.89%, F1-score 91.82%) were reached when the number of features increased to 20. It can be seen that five-fold cross-validation is better than ten-fold cross-validation, which is consistent with the results of the analysis using the confusion matrix in Fig 9. This consistency across multiple evaluation facets solidifies the decision to adopt the 22 features corresponding to five-fold cross-validation as the optimal acoustic source biometric features for subsequent feature fusion. This judicious selection is underpinned by the superior performance and stability exhibited within the five-fold cross-validation framework.

Evaluation of thermal source deep physiological features

This experiment aims to evaluate the capability of different CNN feature extractors in extracting deep physiological features from thermal sources in a given dataset. The optimal deep

热源深度生理特征评估涉及选择最佳特征提取器，用于从热源中提取深度生理特征，并通过将单一深度生理特征输入支持向量机分类器来评估红外数据中的咳嗽分类能力。特征融合评估结合前两项实验中提取的最佳特征，采用不同的融合策略。它通过将融合特征输入支持向量机分类器来评估融合特征的咳嗽分类性能。最后，识别速度和模型大小评估比较并评估前三项实验中选出的最佳方法的识别速度和模型大小。

声源生物特征的评估

本实验旨在评估SVM-RFECV算法的性能，并从29种咳嗽声音特征中根据实验结果选择最优特征集作为声源生物特征。这些选定特征将被输入到支持向量机分类器中，以评估单个声学特征在咳嗽识别中的分类性能。在将SVM-RFECV与其他特征选择方法进行比较时，我们考虑了导致我们选择SVM-RFECV的几个因素。SVM-RFECV提供了一种迭代过程，可以消除无关或冗余的特征，有效减少噪声的影响并增强模型的鲁棒性。此外，SVM-RFECV利用了支持向量机（SVMs）的力量，支持向量机以其处理高维数据并捕捉复杂模式的能力而闻名。我们通过结合SVM与RFECV实现精确的特征选择，同时保持计算效率。这些优势使SVM-RFECV成为我们特定应用的合适选择，并促使其在替代特征选择方法中脱颖而出。

关于特征选择指标，我们采用了F1分数，该分数同时考虑了精确率和召回率，为模型性能提供了全面的评估。通过评估F1分数，我们可以有效地衡量精确率和召回率之间的权衡，确保所选特征优化了分类结果。

在采用SVM-RFECV进行特征选择时，我们评估了模型中的特征及其相应的性能，使用交叉验证测试分数。具体而言，图7和图8分别展示了使用五折和十折交叉验证的评估结果。在图7中，我们观察到当使用八个特征时，F1分数得到提升，并且训练和验证数据的F1分数趋于稳定。随后，随着特征数量的增加，模型的准确率逐渐提高。当特征数量增加到22个时，最佳模型评估指标（准确率92.13%，精确率92.18%，召回率92.05%，F1分数92.10%）得以达到。在图8中，整体趋势与图7一致，当特征数量增加到20个时，最佳模型评估指标（准确率91.84%，精确率91.83%，召回率91.89%，F1分数91.82%）得以达到。可以看出，五折交叉验证优于十折交叉验证，这与图9中使用的混淆矩阵分析结果一致。在多个评估方面的这种一致性，巩固了采用对应于五折交叉验证的22个特征作为后续特征融合的最佳声源生物特征的决定。这一明智的选择得益于五折交叉验证框架中展现出的卓越性能和稳定性。

热源深度生理特征评估

这项实验旨在评估不同CNN特征提取器从给定数据集中的热源中提取深度生理特征的能力。最优的深度

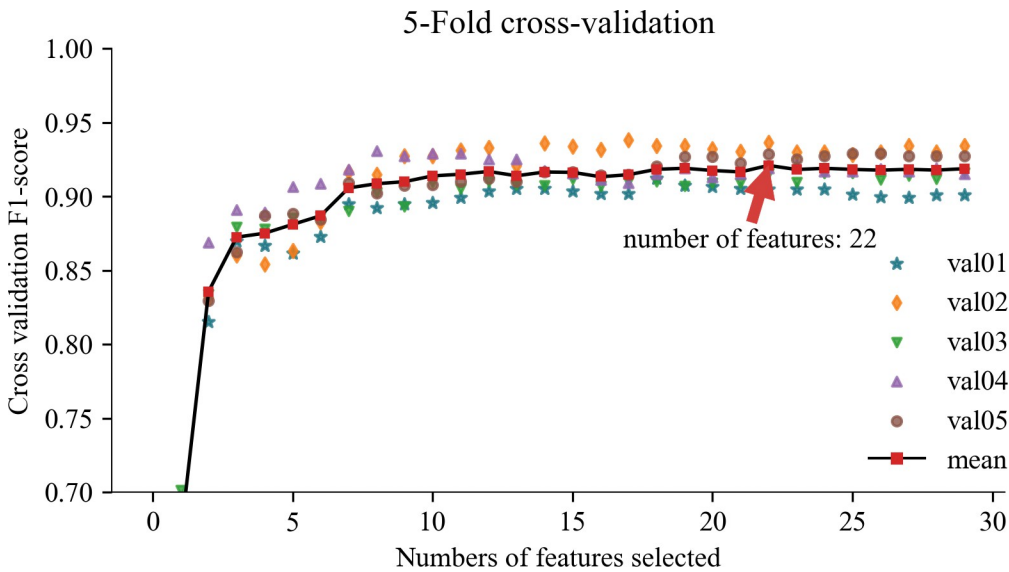


Fig 7. Evaluation results of acoustic source biometric features of 5-Fold.

<https://doi.org/10.1371/journal.pone.0297655.g007>

physiological features obtained from thermal sources will be inputted into an SVM classifier to assess the classification performance of single deep physiological features for cough recognition. The extraction ability of the fully connected layers of Lenet-5, AlexNet, DenseNet121, Vgg16, Vgg19, ResNet50, ResNet101, ResNet152, and ThermographicNet networks for deep physiological features was first evaluated, respectively. An infrared thermal image of the pig cough is taken as input, the most representative CNN network is selected as the feature extractor, and SVM is used as the classifier to accomplish the cough recognition task. Lenet-5 and AlexNet are prominent examples of shallow convolutional neural networks (CNNs), and ThermographicNet is a collection of the most representative CNN networks of Lenet-5 and AlexNet. VGG and ResNet are prominent examples of deep convolutional neural networks

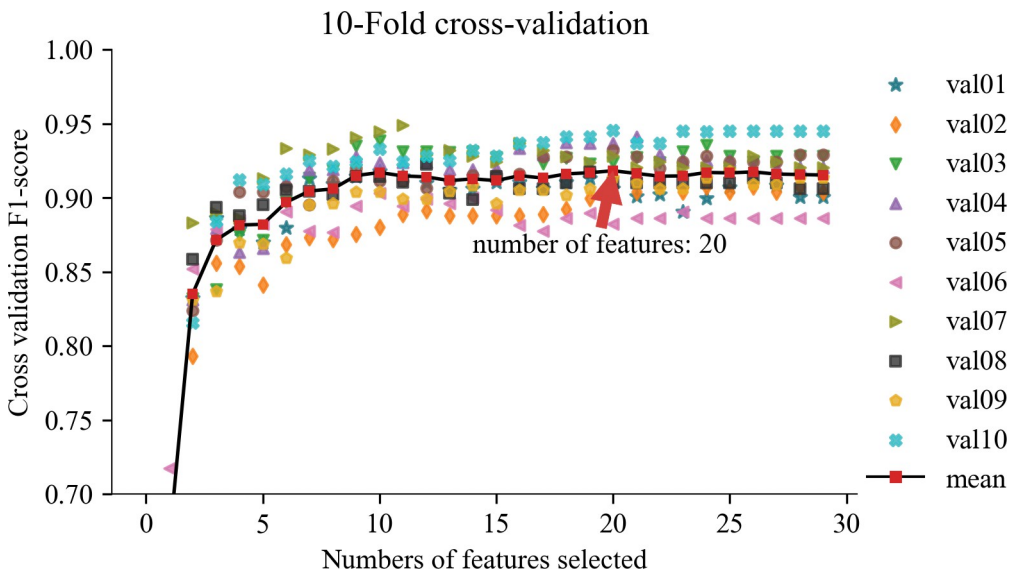


Fig 8. Evaluation results of acoustic source biometric features of 10-Fold.

<https://doi.org/10.1371/journal.pone.0297655.g008>

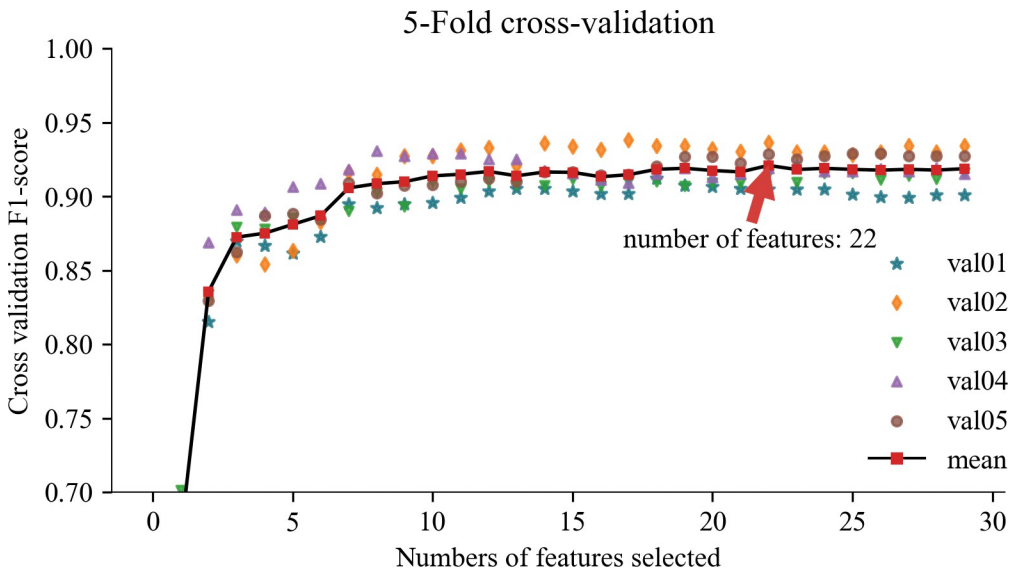


图 7. 声源生物特征的评估结果

5-Fold.

[<https://doi.org/10.1371/journal.pone.0297655.g007>](https://doi.org/10.1371/journal.pone.0297655.g007)

本实验旨在评估不同卷积神经网络特征提取器从给定数据集中热源中提取深度生理特征的能力。从热源获得的最佳深度生理特征将被输入到支持向量机分类器中，以评估单个深度生理特征对咳嗽识别的分类性能。首先分别评估了Lenet-5、AlexNet、DenseNet121、Vgg16、Vgg19、ResNet50、ResNet101、ResNet152和热成像网络对深度生理特征的提取能力。以猪咳嗽的红外热成像为输入，选择最具代表性的卷积神经网络作为特征提取器，并使用支持向量机作为分类器来完成咳嗽识别任务。Lenet-5和AlexNet是浅层卷积神经网络（CNN）的突出例子，而热成像网络是Lenet-5和AlexNet中最具代表性卷积神经网络的集合。VGG和ResNet是深层卷积神经网络的突出例子

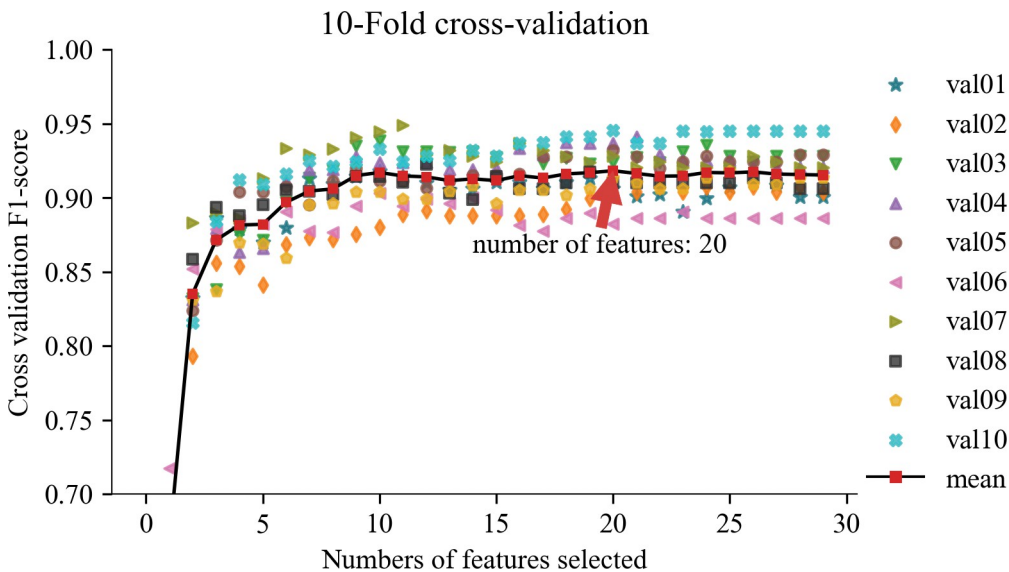


图 8. 声源生物特征的 10-Fold 评估结果

[<https://doi.org/10.1371/journal.pone.0297655.g008>](https://doi.org/10.1371/journal.pone.0297655.g008)

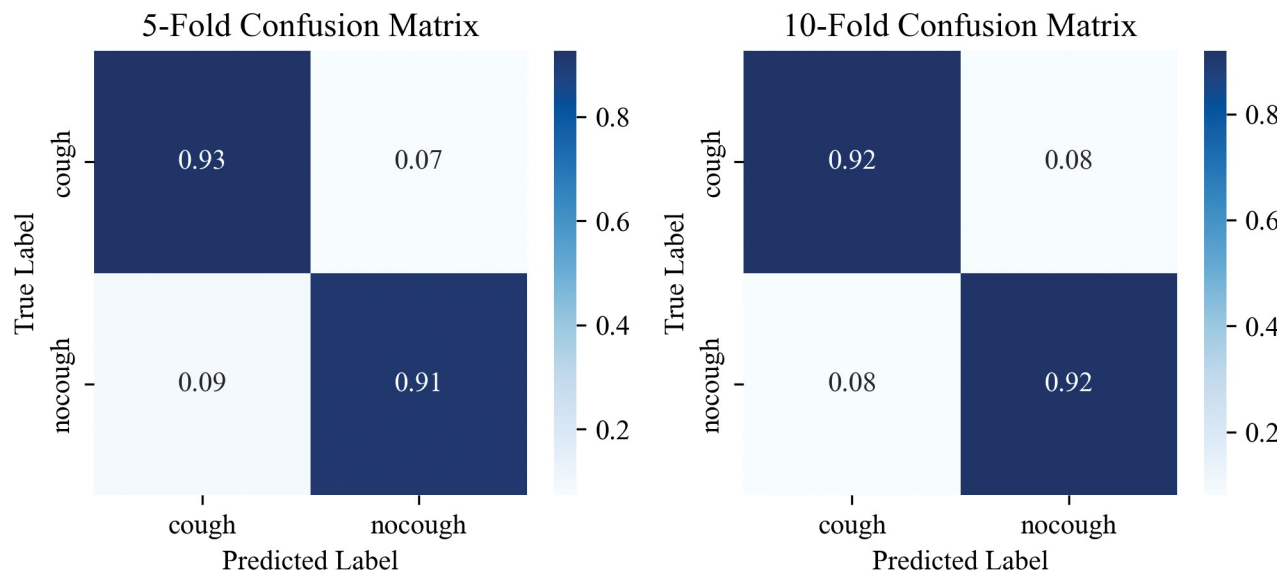


Fig 9. Evaluation results of acoustic source biometric features of confusion matrix results.

<https://doi.org/10.1371/journal.pone.0297655.g009>

(CNNs). Shallow layers are characterized by their small size and portability, making them easy for application deployment. VGG is characterized by emphasizing deep layers with small filters, while ResNet introduces residual connectivity to address the challenge of vanishing gradients in deep networks. These models demonstrate the ability of deep CNNs to capture complex patterns and hierarchical representations. Then, using confusion matrix analysis, the feature vectors with the best performance are selected from the network’s fully connected layers with the most robust feature extraction capability as the thermal source deep physiological features. The performance evaluation metrics, including Accuracy, Precision, Recall, and F1-score, are used to evaluate the classification performance of SVMs using different CNN networks as feature extractors. The experimental results are shown in Table 2, and the corresponding confusion matrix results are shown in Fig 10.

Table 2. Comparative evaluation results of deep physiological features extraction.

CNN Feature Extractor		SVM Classifier Evaluation Metrics			
Name	FC Layer	Accuracy	Precision	Recall	F1-score
Lenet-5	FC1	95.41%	95.45%	95.41%	95.41%
	FC2	95.41%	95.45%	95.41%	95.41%
AlexNet	FC1	96.14%	96.22%	96.14%	96.13%
	FC2	95.82%	95.96%	95.82%	95.82%
DenseNet121	FC1	93.25%	93.33%	93.25%	93.25%
Vgg16	FC1	94.58%	94.63%	94.58%	94.58%
	FC2	94.58%	94.64%	94.58%	94.58%
Vgg19	FC1	93.16%	93.28%	93.16%	93.16%
	FC2	93.40%	93.50%	93.40%	93.39%
ResNet50	FC1	95.99%	96.06%	95.99%	95.99%
ResNet101	FC1	95.13%	95.19%	95.13%	95.12%
ResNet152	FC1	94.0%	94.15%	94.0%	94.0%
ThermographicNet	FC1	96.80%	96.84%	96.80%	96.80%
	FC2	96.68%	96.73%	96.68%	96.68%

<https://doi.org/10.1371/journal.pone.0297655.t002>

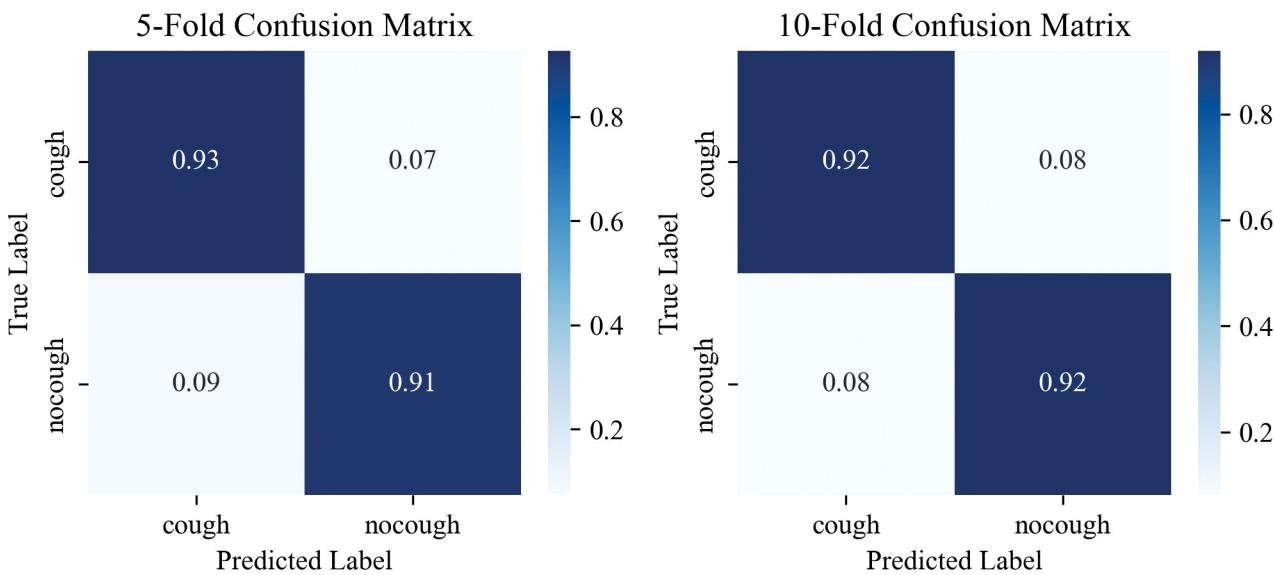


图9. 评估结果 声源生物特征的混淆矩阵结果 矩阵结果。

<<https://doi.org/10.1371/journal.pone.0297655.g009>>

(卷积神经网络). 浅层特征在于其规模小和便携性，便于应用部署。VGG 特征在于强调使用小滤波器构建深层，而 ResNet 引入残差连接以解决深度网络中梯度消失的挑战。这些模型展示了深度卷积神经网络捕获复杂模式和层次化表示的能力。随后，通过混淆矩阵分析，从网络中具有最强特征提取能力的全连接层中选取性能最优的特征向量作为热源深度生理特征。性能评估指标，包括准确率、精确率、召回率和 F1 分数，用于评估使用不同卷积神经网络作为特征提取器时 SVM 的分类性能。实验结果如表2所示，相应的混淆矩阵结果如图10所示。表2，图10。

表2. 深度生理特征提取的比较评估结果

CNN特征提取器		支持向量机分类器评估指标			
Name	全连接层	准确率	精确率	召回率	F1分数
LeNet-5	FC1	95.41%	95.45%	95.41%	95.41%
	FC2	95.41%	95.45%	95.41%	95.41%
AlexNet	FC1	96.14%	96.22%	96.14%	96.13%
	FC2	95.82%	95.96%	95.82%	95.82%
DenseNet121	FC1	93.25%	93.33%	93.25%	93.25%
Vgg16	FC1	94.58%	94.63%	94.58%	94.58%
	FC2	94.58%	94.64%	94.58%	94.58%
Vgg19	FC1	93.16%	93.28%	93.16%	93.16%
	FC2	93.40%	93.50%	93.40%	93.39%
ResNet50	FC1	95.99%	96.06%	95.99%	95.99%
ResNet101	FC1	95.13%	95.19%	95.13%	95.12%
ResNet152	FC1	94.0%	94.15%	94.0%	94.0%
热成像网络	FC1	96.80%	96.84%	96.80%	96.80%
	FC2	96.68%	96.73%	96.68%	96.68%

<<https://doi.org/10.1371/journal.pone.0297655.t002>>

The experimental results in Table 2 show notable differences between different feature extractors for cough classification. The average accuracy and F1-score of shallow CNN network feature extractors (LeNet-5, AlexNet, DenseNet121) are 95.21% and 95.20%, respectively. In contrast, the average accuracy and F1-score of deep CNN network feature extractors (Vgg16, Vgg19, ResNet50, ResNet101, ResNet152) are 94.41% and 94.40%, respectively. Our proposed ThermographicNet network feature extractor achieves an average accuracy and F1-score of 96.74%.

From the confusion matrix results in Fig 10, it is evident that, except for VGG19, the recognition performance of the fully connected layers in other network architectures is similar. FC1

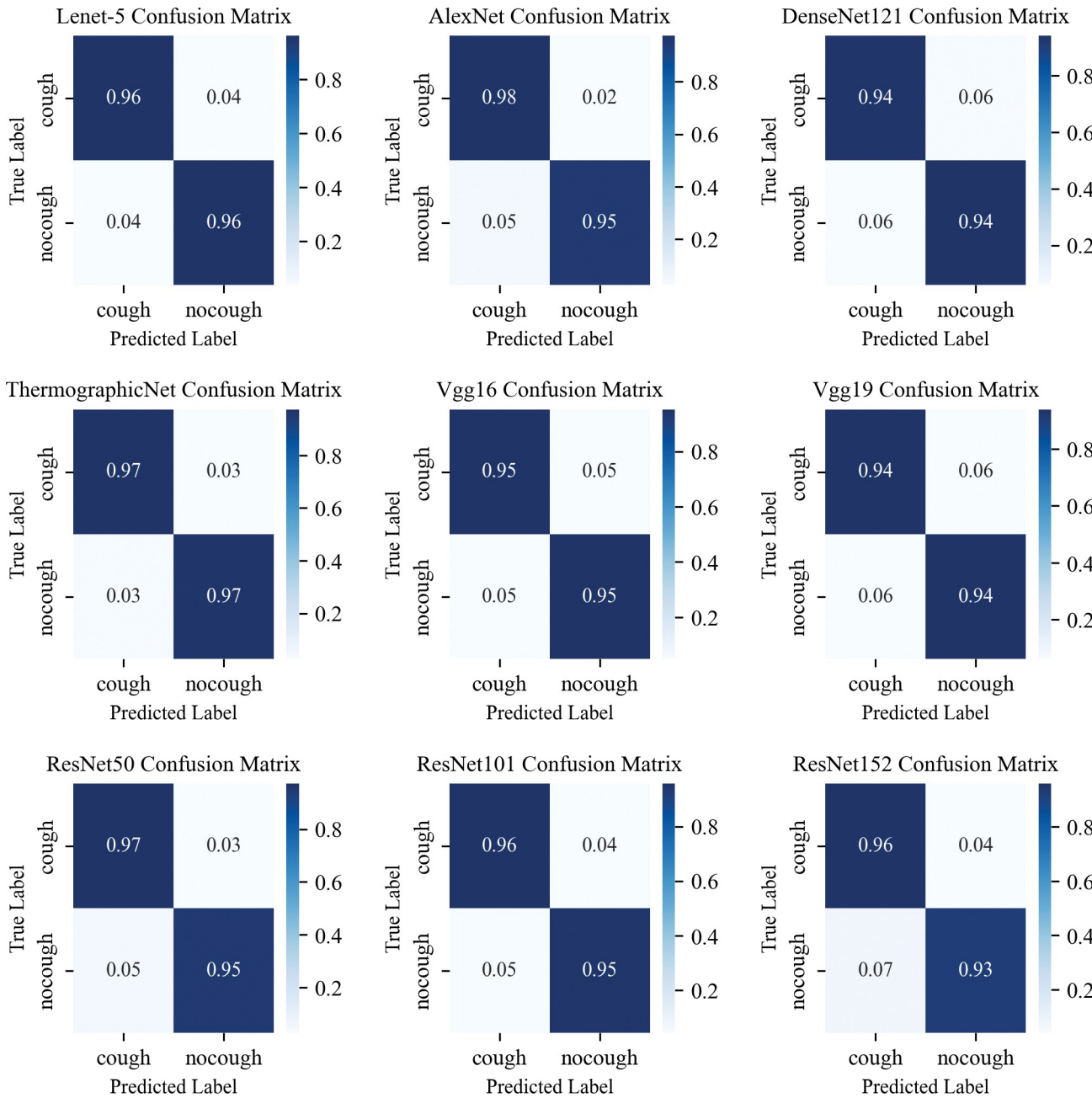


Fig 10. Confusion matrix results of thermal source deep physiological features.

<https://doi.org/10.1371/journal.pone.0297655.g010>

实验结果在表2中显示了不同咳嗽分类特征提取器之间的显著差异。浅层卷积神经网络特征提取器（LeNet-5、AlexNet、DenseNet121）的平均准确率和F1分数分别为95.21%和95.20%，而深层卷积神经网络特征提取器（Vgg16、Vgg19、ResNet50、ResNet101、ResNet152）的平均准确率和F1分数分别为94.41%和94.40%。我们提出的ThermographicNet网络特征提取器实现了96.74%的平均准确率和F1分数。

从混淆矩阵结果在图10中可以看出，除了Vgg19，其他网络架构中的全连接层识别性能相似。FC1

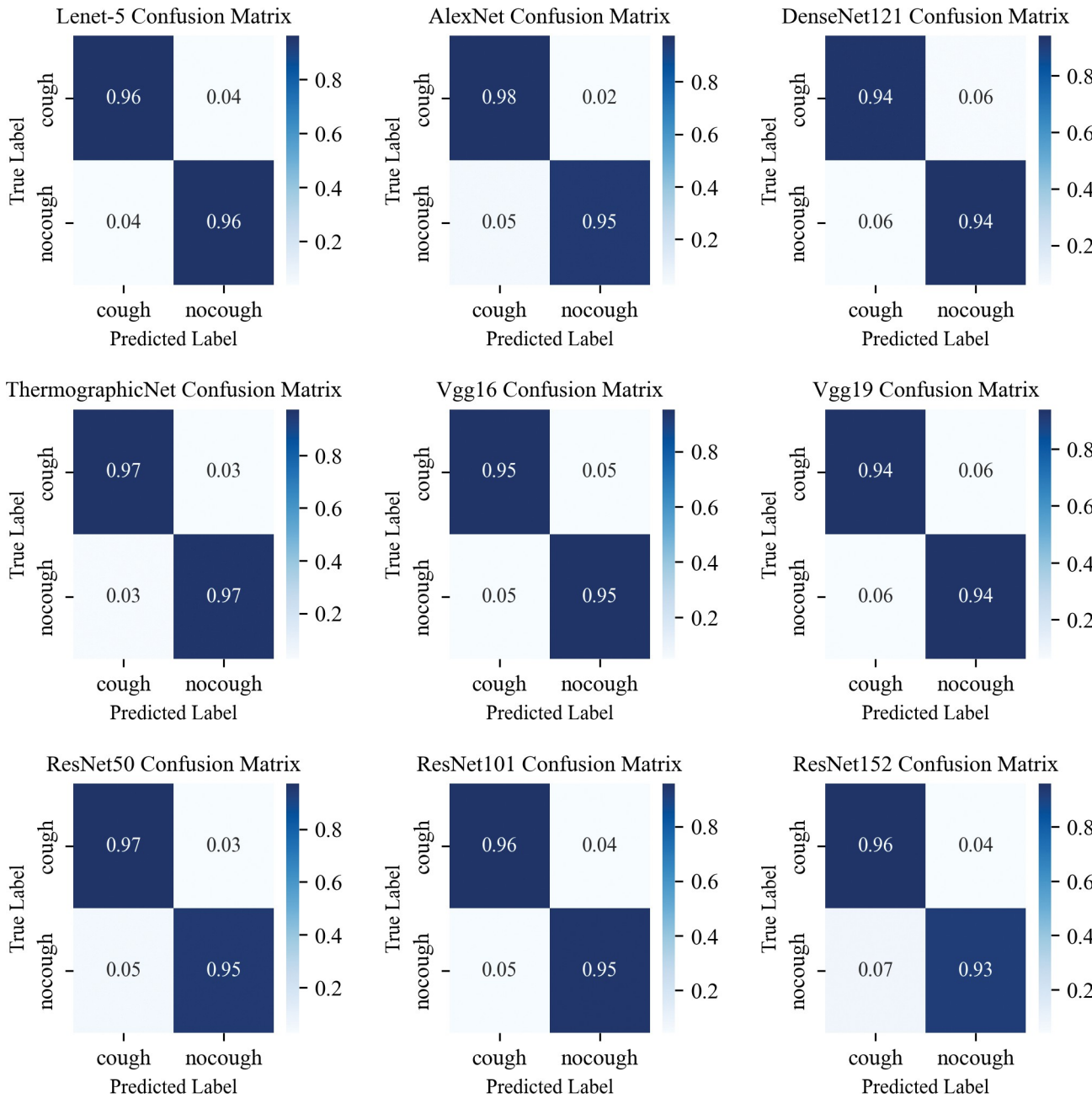


图10. 热源深度生理特征的混淆矩阵结果。

<https://doi.org/10.1371/journal.pone.0297655.g010>

Table 3. Comparative analysis of pig cough recognition using different fusion strategies.

Features/Fusion features	SVM Classifier Evaluation Metrics			
	Accuracy	Precision	Recall	F1-score
acoustic source biometric features	92.13%	92.18%	92.05%	92.10%
FC1 thermal source deep physiological features	96.80%	96.84%	96.80%	96.80%
FC2 thermal source deep physiological features	96.68%	96.73%	96.68%	96.68%
layer fusion features	96.86%	96.89%	96.86%	96.86%
FC1 heterogeneous fusion features	98.79%	98.80%	98.79%	98.79%
FC2 heterogeneous fusion features	98.67%	98.68%	98.67%	98.67%
Layer heterogeneous fusion features	98.62%	98.62%	98.62%	98.62%

<https://doi.org/10.1371/journal.pone.0297655.t003>

consistently outperforms FC2 in classification performance. Furthermore, the Thermogra-
phicNet network proposed in this study demonstrates the best recognition performance across
all four evaluation metrics. Based on the experimental results, we chose ThermographicNet
network as the deep feature extractor and two fully connected layer features as the thermal
source deep physiological features.

Evaluation of feature fusion

This experiment aimed to evaluate the ability of four different fusion strategies to classify pig
coughing behavior to select the optimal fusion classification strategy. In Evaluation of acoustic
source biometric features Section, the experiment selected the best acoustic set of 22 sound fea-
tures as the acoustic source biometric features. In Evaluation of thermal source deep physio-
logical features Section, the experiment was validated using our proposed ThermographicNet
network as the feature extractor, obtaining the optimal classification performance based on the
feature vectors from the FC1 layer’s output, which served as the thermal source deep physio-
logical features. Following the fusion methods described in Feature fusion methods Section,
the performance of four fusion methods, namely layer fusion, FC1 heterogeneous fusion, FC2
heterogeneous fusion, and layer heterogeneous fusion, were evaluated for cough classification,
as presented in Table 3. The confusion matrix analysis results in Fig 11 demonstrate that differ-
ent fusion strategies yield better results than a single feature. Three heterogeneous fusion
methods outperform the homogeneous fusion method significantly. It is not surprising that
combining biometric features with physiological features proves to be a practical approach,
significantly improving cough recognition performance. The FC1 heterogeneous fusion
method achieves an impressive classification accuracy and F1-score of 98.79%.

The three aforementioned experiments validate the proposed method as the optimal
approach for cough recognition. Initially, the SVM-RFECV algorithm was employed to extract
22 features as acoustic source biometric features, which were then fed into an SVM classifier to
evaluate the performance of using a single acoustic feature for cough recognition. Subse-
quently, the FC1 layer of the ThermographicNet network was utilized as a feature extractor to
extract deep physiological features from thermal sources. These features were then evaluated
using an SVM classifier to assess the performance of using thermal data alone for cough recog-
nition. Finally, an early fusion technique combined the representative and significant features
from both acoustic and thermal sources. The fused features were then inputted into an SVM
classifier to evaluate the ability of heterogeneous fusion features in cough recognition. The
experimental results demonstrate that the fused features outperform any single feature, thereby
confirming the superiority and feasibility of the proposed method.

表3. 不同融合策略的猪咳嗽识别对比分析。

特征/融合特征	SVM分类器评估指标			
	准确率	精确率	召回率	F1分数
声源生物特征	92.13%	92.18%	92.05%	92.10%
FC1 热源深度生理特征	96.80%	96.84%	96.80%	96.80%
全连接层2热源深度生理特征	96.68%	96.73%	96.68%	96.68%
层融合特征	96.86%	96.89%	96.86%	96.86%
FC1异构融合特征	98.79%	98.80%	98.79%	98.79%
FC2异构融合特征	98.67%	98.68%	98.67%	98.67%
层异构融合特征	98.62%	98.62%	98.62%	98.62%

<<https://doi.org/10.1371/journal.pone.0297655.t003>>

在分类性能方面始终优于FC2。此外，本研究提出的ThermalographicNet网络在所
有四个评估指标中均表现出最佳识别性能。基于实验结果，我们选择
ThermalographicNet网络作为深度特征提取器，并将两个全连接层特征作为热源深度
生理特征。

特征融合评估

本实验旨在评估四种不同融合策略对猪咳嗽行为的分类能力，以选择最优的融合分类
策略。在声源生物特征评估部分，实验选取了最佳的22个声音特征作为声源生物特征。
在热源深度生理特征评估部分，实验使用我们提出的ThermographicNet网络作为特
征提取器进行验证，基于FC1层的输出特征向量，获得了最优的分类性能，将其作为
热源深度生理特征。根据特征融合方法部分所述的融合方法，对四种融合方法（即层
融合、FC1异构融合、FC2异构融合和层异构融合）的咳嗽分类性能进行了评估，结
果如表3所示表3。混淆矩阵分析结果在图11中显示，不同的融合策略比单一特征能获得
更好的结果。三种异构融合方法显著优于同构融合方法。将生物特征与生理特征相
结合是一种实用的方法，显著提高了咳嗽识别性能。FC1异构融合方法实现了令人印
象深刻的分类准确率和F1分数，达到98.79%。

前述三个实验验证了所提出的方法是咳嗽识别的最佳方法。最初，采用
SVM-RFECV算法提取22个声源生物特征，然后将这些特征输入到SVM分类器中，以
评估使用单个声学特征进行咳嗽识别的性能。随后，利用ThermographicNet网络的
FC1层作为特征提取器，提取深层生理特征。

来自热源的特征。然后使用支持向量机分类器评估这些特征，以评估仅使用热数
据进行咳嗽识别的性能。最后，一种早期融合技术结合了来自声学和热源的代表性及
显著特征。融合后的特征被输入到支持向量机分类器中，以评估异构融合特征在咳嗽
识别中的能力。实验结果表明，融合特征优于任何单一特征，从而证实了所提出方法
的优越性和可行性。

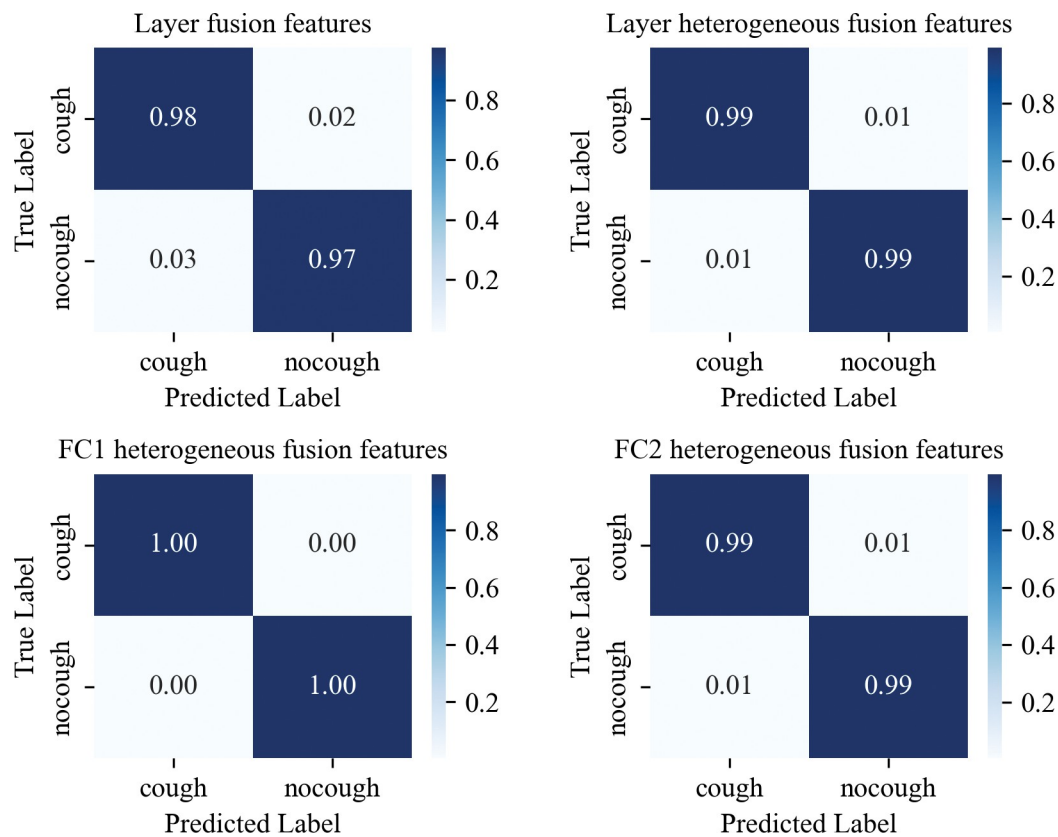


Fig 11. Confusion matrix results of fusion feature.

<https://doi.org/10.1371/journal.pone.0297655.g011>

Evaluation of recognition speed and model size

To further affirm the practical applicability of our proposed method in enhancing animal welfare, a comprehensive evaluation of the model’s recognition speed and size is imperative. An effective classification model should not only exhibit superior classification performance but also excel in recognition speed and model size, both of which are pivotal factors influencing the real-world deployment of the system. In field situations, computational efficiency and model size significantly impact deployment conditions, environmental settings, and overall costs. It is crucial to minimize computation time and achieve runtime acceleration for an efficient recognition process [48]. Therefore, we conducted a comparative assessment, evaluating ThermographicNet against other CNN feature extractors for deployment applications.

A set of samples (5 samples) was randomly selected from our test set, each of which was computed five times on the computing platform without any additional load. The average execution time per sample and the average execution time per group were counted. The results are shown in Fig 12, a solid circle is the average time, and a hollow diamond is the individual time. From the experimental results. Our proposed Heterogeneous fusion method using ThermographicNet as a feature extractor has the fastest processing time, leading to a rapid increase in recognition speed. On average, the execution time of our proposed method is comparable to shallow CNNs, faster than all other deep CNNs, 4.7 times faster than the VGG family, and 13.9 times faster than the ResNet family.

Finally, the model sizes are evaluated because they directly affect the deployment scenario [49], and the results are shown in Fig 13. Compared to the other eight models, the size of our

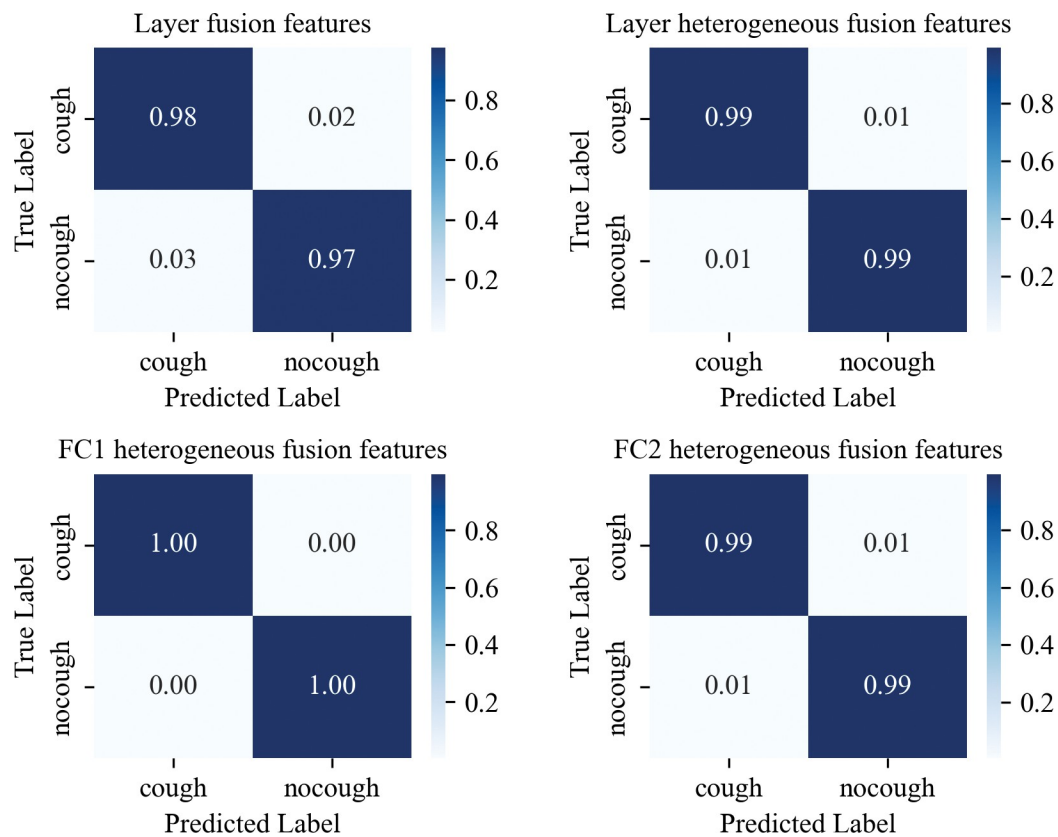


图 11. 融合特征的混淆矩阵结果。

<https://doi.org/10.1371/journal.pone.0297655.g011>

识别速度和模型大小评估

为了进一步证实我们提出的方法在提升动物福利方面的实际应用价值，对模型的识别速度和大小进行全面评估至关重要。一个有效的分类模型不仅应表现出优异的分类性能，还应擅长识别速度和模型大小，这两者都是影响系统实际部署的关键因素。在实际场景中，计算效率和模型大小会显著影响部署条件、环境设置和总体成本。为了实现高效的识别过程，必须最小化计算时间并实现运行时加速 [48]。因此，我们进行了对比评估，将ThermographicNet与其他CNN特征提取器在部署应用方面进行了比较。

从我们的测试集中随机选择了一组样本（5个样本），在计算平台上对每个样本进行了五次计算，没有任何额外的负载。计算了每个样本的平均执行时间和每个组的平均执行时间。结果如图12所示，实心圆圈是平均时间，空心菱形是个体时间。从实验结果来看。我们提出的采用ThermographicNet作为特征提取器的异构融合方法具有最快的处理时间，从而显著提高了识别速度。平均而言，我们提出的方法的执行时间与浅层CNN相当，比所有其他深度CNN都快，比VGG系列快4.7倍，比ResNet系列快13.9倍。

最后，评估模型大小，因为它们直接影响部署场景[49],和结果展示在图13。与其他八个模型相比，我们

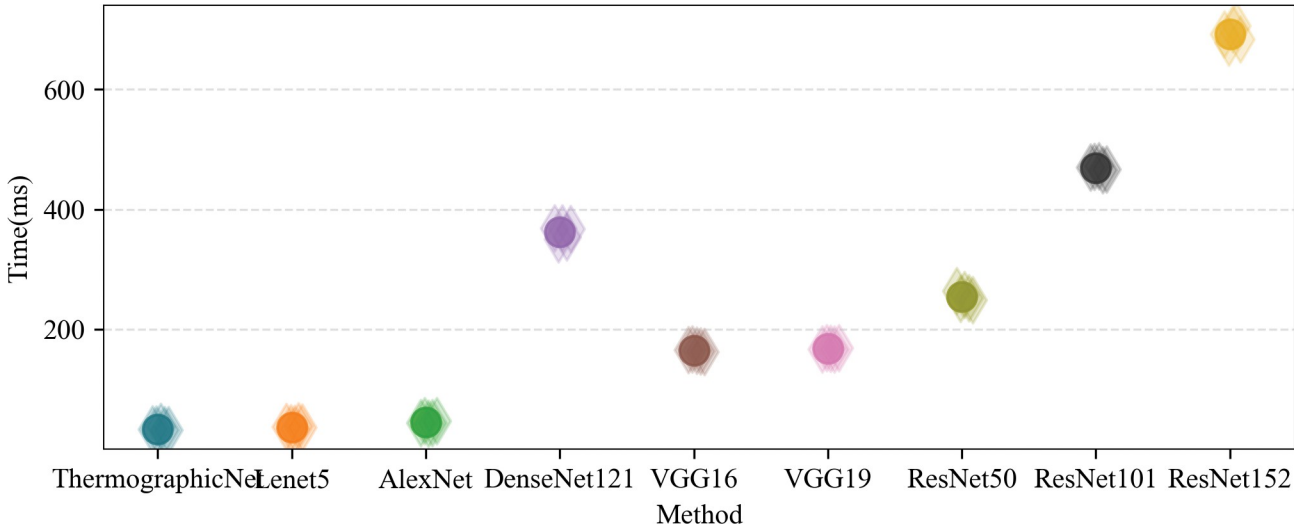


Fig 12. Comparison of computational time with different CNN networks using point plots.

<https://doi.org/10.1371/journal.pone.0297655.g012>

model is comparable to the size of the shallow series model, our model size is 12.5% of the average size of the VGG family and less than 2.7% of the ResNet family. Therefore, in terms of model size cost, our model has a clear deployment advantage. It can be deployed and executed not only on central servers and edge servers but also on portable devices, enhancing its versatility and accessibility.

Discussion

According to Figs 7 and 8, the acoustic source biometric feature set exhibits satisfactory results in pig cough recognition, but there still needs to be more potential for performance improvement. The curve indicates a trend of feature elimination, suggesting that incorporating additional features could enhance classification performance and expand the dimensionality of acoustic features [50,51]. However, the slope of the curve in Figs 7 and 8 implies that the

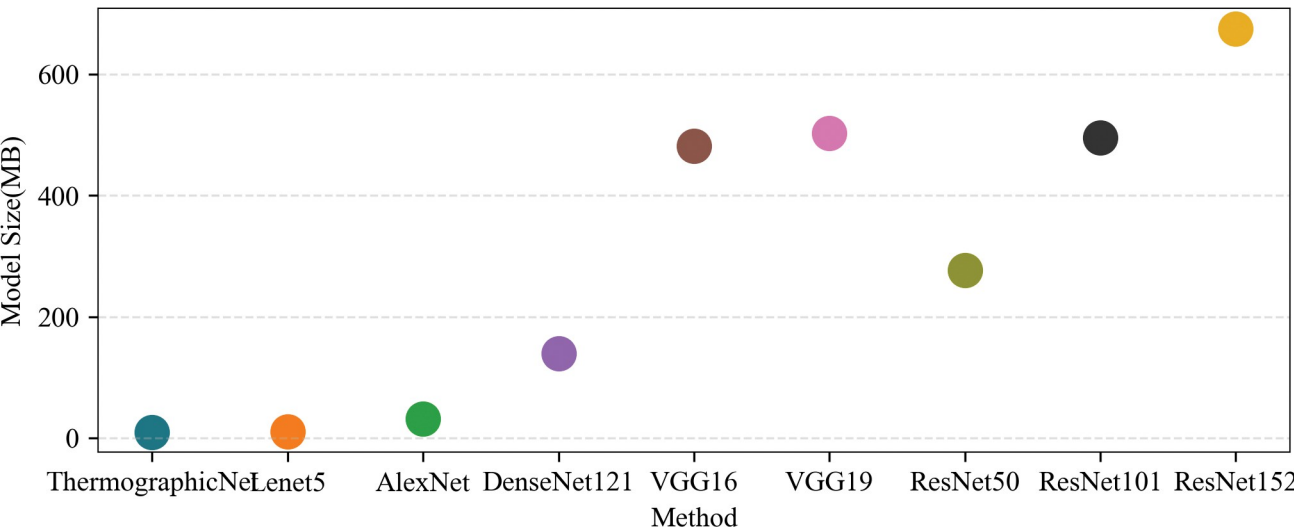


Fig 13. Comparison of model size with different CNN networks using point plots.

<https://doi.org/10.1371/journal.pone.0297655.g013>

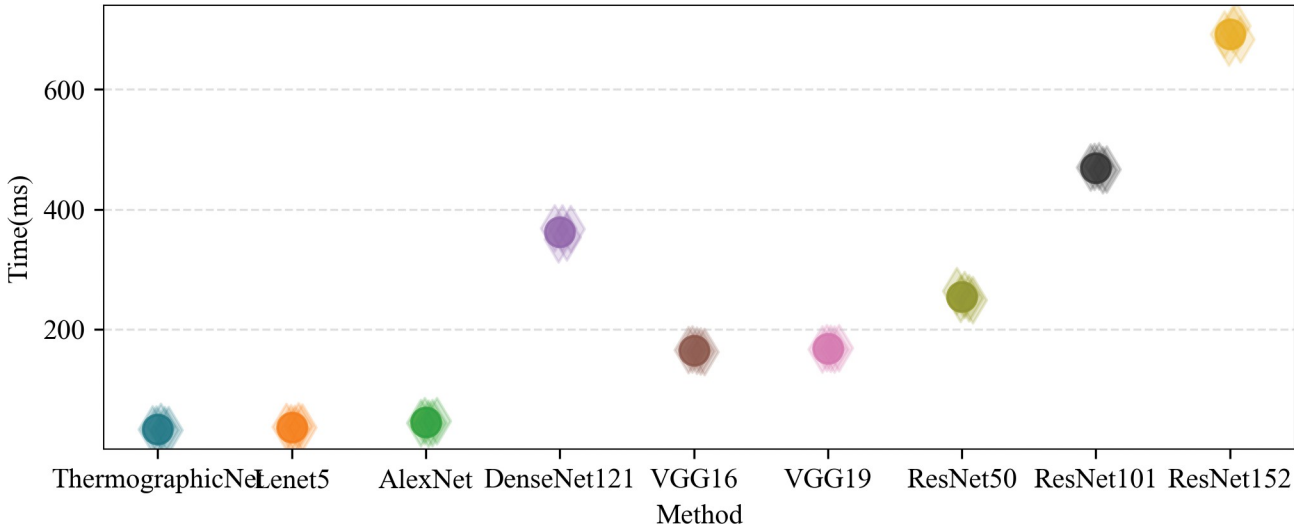


图 12. 使用点图比较不同卷积神经网络的计算时间。

[<https://doi.org/10.1371/journal.pone.0297655.g012>](https://doi.org/10.1371/journal.pone.0297655.g012)

该模型的大小与浅层系列模型相当，我们的模型大小是VGG家族平均大小的12.5%，不到ResNet家族的2.7%。因此，在模型大小成本方面，我们的模型具有明显的部署优势。它不仅可以部署和执行在中心服务器和边缘服务器上，还可以部署在便携式设备上，增强了其通用性和可访问性。

讨论

根据图 7 和 8，声源生物特征集在猪咳嗽识别中表现出令人满意的结果，但性能仍有提升空间。曲线显示特征消除的趋势，表明增加更多特征可以提高分类性能并扩展声学特征的维度 [50,51]。然而，图 7 和 8 中的曲线斜率表明

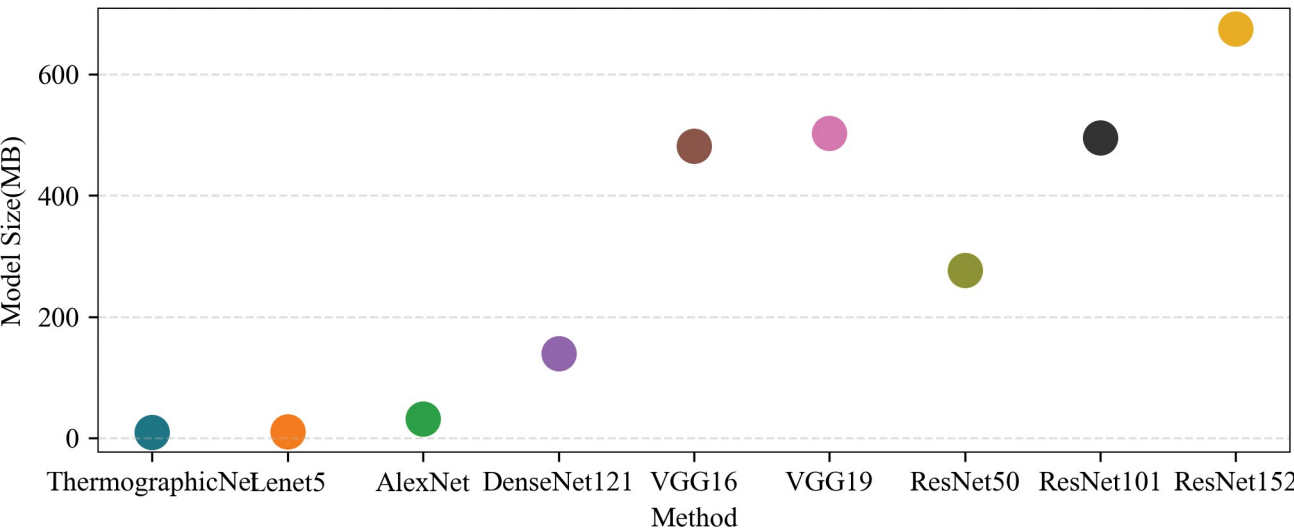


图 13. 使用点图比较不同卷积神经网络模型的模型大小。

[<https://doi.org/10.1371/journal.pone.0297655.g013>](https://doi.org/10.1371/journal.pone.0297655.g013)

performance gains from adding homogeneous acoustic features are increasingly challenging. To address this, we can enhance performance by exploring more diverse extended features and selecting a smaller, more refined feature set tailored specifically for pig cough recognition. Future research should explore additional features beyond the three domains of acoustics. Furthermore, extracting deep features from sound can enhance the recognition performance of individual audio samples [52], thereby determining the optimal set of acoustic features for pig cough sound recognition. Additionally, it is crucial to consider incorporating signal features beyond acoustics to overcome the performance bottleneck associated with single-feature analysis. Existing studies support these findings. For instance, Samson et al. [53] demonstrated that selecting representative features significantly improves classification accuracy in speech emotion recognition. Similarly, Saisanthiya et al.[54]and Raheel et al. [55] showcased the performance enhancement achieved by integrating heterogeneous features such as text or physiological signals in sound recognition tasks. Kakuba et al. [56] conducted an extensive exploration of modal and cross-modal modeling machine fusion methods, focusing on fusion studies involving sound and lexical semantics across various temporal, spatial, and semantic dimensions. Drawing inspiration from this work, future investigations into animal behavior could leverage a similar approach by integrating different modalities, including behavior, physiology, ecology, and genomics. This holistic perspective promises more practical and applied research outcomes in the realm of animal behavior studies. In summary, by leveraging a diverse range of extended features and carefully selecting a refined feature set, we can further enhance the performance of pig cough recognition. Future research should continue expanding acoustic features while considering incorporating additional signal features.

The experimental results presented in Table 2 clearly demonstrate the significant potential of thermal image depth features as a valuable tool for identifying pig coughs under field conditions. Infrared images have proven to be remarkably successful in recognizing cough patterns [57,58], with performance on par with current mainstream research focused on sound-based cough recognition. Notably, in the shallow CNN feature extractor, AlexNet (FC1) exhibited impressive performance, achieving an F1 score of 96.13%. Similarly, the deep CNN feature extractor, ResNet50 (FC1), demonstrated an outstanding performance with an impressive score of 95.99%. Evidently, shallow CNNs outperformed their deep counterparts in extracting essential features, while the inclusion of a fully connected layer had a limited impact on enhancing the classification performance. This outcome can be attributed to the characteristics of thermal image data, where shallow CNNs already capture a sufficient amount of information. This finding is in line with Ji et al. [16], which favours us to improve our model to obtain better future results. This indirectly validates the rationale behind leveraging the strengths of Lenet and AlexNet to design ThermographicNet, a deep physiological feature extractor tailored for thermal sources, as proposed in this paper. Additionally, Figs 12 and 13 further validate the superiority of our feature extractor in terms of its ability to extract features, processing speed, and model size.

The results presented in Table 3 demonstrate that the fusion of acoustic source biometric features and thermal source physiological deep features in a multi-source framework is an efficient and robust approach for pig cough recognition. However, when it comes to the homogeneous fusion of deep physiological features using the Layer fusion method, only a 96.86% F1 score was achieved. This is only slightly better than the 96.8% F1 score obtained with a singlethermal source physiological deep feature. Moreover, adding more homogeneous features and increasing computation and storage complexity did not significantly improve the classification performance. These findings align with previous research on homogeneous fusion in the field of speech [16,17], indicating that there is a performance bottleneck in homogeneous fusion.

增加同质声学特征带来的性能提升正变得越来越具有挑战性。为了应对这一问题，我们可以通过探索更多样化的扩展特征，并选择一个更小、更精炼的、专门针对猪咳嗽识别的特征集来提升性能。未来的研究应该探索超出三个声学域之外的其他特征。此外，从声音中提取深度特征可以增强单个音频样本的识别性能 [52],从而确定猪咳嗽声音识别的最佳声学特征集。此外，考虑结合声学以外的信号特征以克服单特征分析的性能瓶颈至关重要。现有研究支持这些发现。例如，Samson等人 [53] 证明了选择代表性特征可以显著提高语音情感识别的分类准确率。类似地，Saisanthiya等人[54]和Raheel等人 [55]展示了通过在声音识别任务中整合文本或生理信号等异构特征所实现的性能提升。Kakuba等人 [56]对模态与跨模态建模机器融合方法进行了广泛的探索，重点关注涉及声音与词汇语义的融合研究，涵盖各种时间、空间和语义维度。受此工作的启发，未来的动物行为研究可以通过整合不同模态（包括行为、生理学、生态学和基因组学）来借鉴类似方法。这种整体视角有望在动物行为研究领域带来更实用、更应用的研究成果。总之，通过利用多样化的扩展特征并仔细选择精炼的特征集，我们可以进一步提升猪咳嗽识别的性能。未来的研究应继续扩展声学特征，同时考虑结合其他信号特征。

实验结果如表2所示，清晰地表明热成像深度特征作为识别田间猪咳嗽的有效工具具有显著潜力。红外图像已被证明在识别咳嗽模式方面表现出色，性能与当前主流方法相当[57,58],

——基于声音的咳嗽识别是主流研究方向。值得注意的是，在浅层CNN特征提取器中，AlexNet (FC1) 表现出令人印象深刻的性能，F1分数达到96.13%。同样，深层CNN特征提取器ResNet50 (FC1) 也展现出卓越的性能，F1分数高达95.99%。显然，浅层CNN在提取关键特征方面优于其深层对应物，而全连接层的加入对提升分类性能的影响有限。这一结果可归因于热成像数据的特性——浅层CNN已能捕捉足够的信息。这一发现与Ji等人的研究一致 [16],，这促使我们改进模型以获得更好的未来结果。这间接验证了本文提出的利用Lenet和AlexNet的优势设计ThermographicNet——一个专为热源设计的深度生理特征提取器——的合理性。此外，图12和图13进一步验证了我们的特征提取器在特征提取能力、处理速度和模型大小方面的优越性

表3中展示的结果表明，在多源框架下融合声源生物特征和热源生理深度特征是一种高效且鲁棒的猪咳嗽识别方法。然而，当使用层融合方法进行深度生理特征的异构融合时，仅达到了96.86%的F1分数。这仅略优于使用单一热源生理深度特征获得的96.8%的F1分数。此外，增加更多异构特征并提高计算和存储复杂度并未显著提升分类性能。这些发现与语音领域关于异构融合的先前研究一致，表明异构融合存在性能瓶颈。 [16,17],这表明异构融合存在性能瓶颈。

On the other hand, three heterogeneous fusion methods, namely FC1 heterogeneous fusion, FC2 heterogeneous fusion, and layer heterogeneous fusion, demonstrated notable performance improvements and validated the complementary nature of heterogeneous features. Particularly, the FC1 heterogeneous fusion method achieved an astonishing 98.79% classification performance, which currently stands as the best result in cough recognition through feature fusion. This can be attributed to the fact that the shallow abstraction depth features derived from infrared data already effectively represent the physiological depth features of cough. Further, adding fully connected layers and fusing more homogeneous feature data contributed little to the classification. Conversely, fusing heterogeneous sound and infrared feature data, which represent biological and physiological signals, respectively, proved to be mutually beneficial and complementary. This insight suggests that future research on animal behavior recognition should consider a multimodal perception approach, incorporating tactile, auditory, and visual modalities.

In addition to achieving successful classification performance, the Heterogeneous fusion method proposed in this study demonstrated clear advantages in model size and recognition speed, as evidenced by the experimental results in Figs 12 and 13. This makes it suitable for deployment in real farm production environments, thereby improving animal welfare and enhancing early clinical diagnostic applications, especially for monitoring respiratory health in pigs.

Lastly, a recent analysis of the latest research on porcine cough recognition revealed that Yin et al. [14] achieved a high score of 99.2% by incorporating late fusion, specifically classifier fusion, in addition to feature fusion. This suggests that, in certain scenarios where time and space complexity requirements are not stringent, other fusion techniques beyond feature fusion can be considered to enhance cough recognition performance. It presents a potential research direction to explore.

One of the main challenges in training deep learning networks is the requirement for large amounts of labeled data, and a potential future solution could be adopting a weakly supervised learning paradigm [59]. Another challenge lies in the interpretability of the model, where the focus on the right features becomes more crucial than the overall accuracy. To delve into the interpretability of analyzing infrared physiological features, we employed the Thermographic-Net model constructed in this study for an interpretability analysis using the Gradient Weighted Class Activation Mapping (Grad-CAM) technique. Eight infrared images of coughing pigs were randomly selected from the test set, and the resulting Grad-CAM maps are presented in Fig 14. As evident from Fig 14, the depth feature sites extracted by the

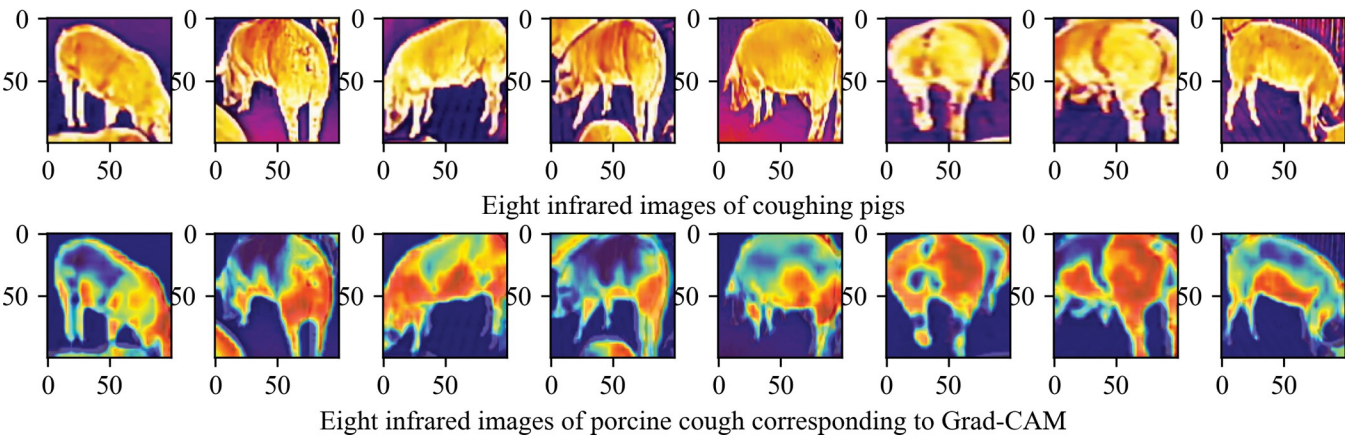


Fig 14. Gradient weighted class activation mapping of ThermographicNet.

<https://doi.org/10.1371/journal.pone.0297655.g014>

另一方面，三种异构融合方法，即FC1异构融合、FC2异构融合和层异构融合，展现了显著的性能提升，并验证了异构特征的互补性。特别是FC1异构融合方法实现了惊人的98.79%分类性能，目前是特征融合咳嗽识别的最佳结果。这可归因于从红外数据中提取的浅层抽象深度特征已能有效表示咳嗽的生理深度特征。此外，增加全连接层并融合更多同构特征数据对分类贡献不大。相反，融合代表生物和生理信号（分别）的异构声音和红外特征数据被证明是互利互补的。这一见解表明，未来的动物行为识别研究应考虑多模态感知方法，结合触觉、听觉和视觉模态。

除了实现成功的分类性能外，本研究提出的异构融合方法在模型大小和识别速度方面表现出明显优势，实验结果在图12和图13中得到证实。这使得它适用于实际农场生产环境，从而改善动物福利并增强早期临床诊断应用，特别是用于监测猪的呼吸健康。

最后，最近对猪咳嗽识别最新研究的分析表明，Yin等人 [14]通过结合后期融合（特别是分类器融合）和特征融合，实现了99.2%的高分。这表明，在时间和空间复杂度要求不严格的情况下，可以考虑除特征融合之外的其他融合技术来提高咳嗽识别性能。这为探索潜在的研究方向提供了参考。

训练深度学习网络的主要挑战之一是需要大量标记数据，而未来的潜在解决方案可能是采用弱监督学习范式 [59]。另一个挑战在于模型的可解释性，即

关注右侧特征变得比整体准确率更为关键。为了深入分析红外生理特征的 interpretability，我们采用本研究构建的 Thermographic-Net 模型，并使用梯度加权类激活映射 (Grad-CAM) 技术进行可解释性分析。从测试集中随机选取了八张咳嗽猪的红外图像，得到的 Grad-CAM 图像展示在图 14。从图 14 中可以看出，ThermographicNet 模型提取的深度特征位点主要集中在猪的口、鼻、耳和腹股沟部位。这些关键位点从动物生理学和病理学的角度来看完全可解释。由上呼吸道感染引起的咳嗽主要表现为耳道、口、鼻和腹股沟部位温度升高。这在红外数据中得以反映，这些特定部位的红外特征更为突出，而我们的特征提取器能够巧妙地捕捉这些特征。因此，不仅可行使用深度特征进行自动呼吸健康检测的研究，而且对猪而言具有生理可解释性。这构成了本文的一项主要贡献。此外，从热红外图像中提取生理深度特征可能需要额外的设备、技术支持以及更复杂的数据采集环境。这可能涉及采购热红外成像设备并实施复杂的图像处理和分析技术，从而增加研究难度和成本，并限制其适用性和泛化性。因此，在实际部署中，应充分考虑农场现有监控和其他系统的综合应用。

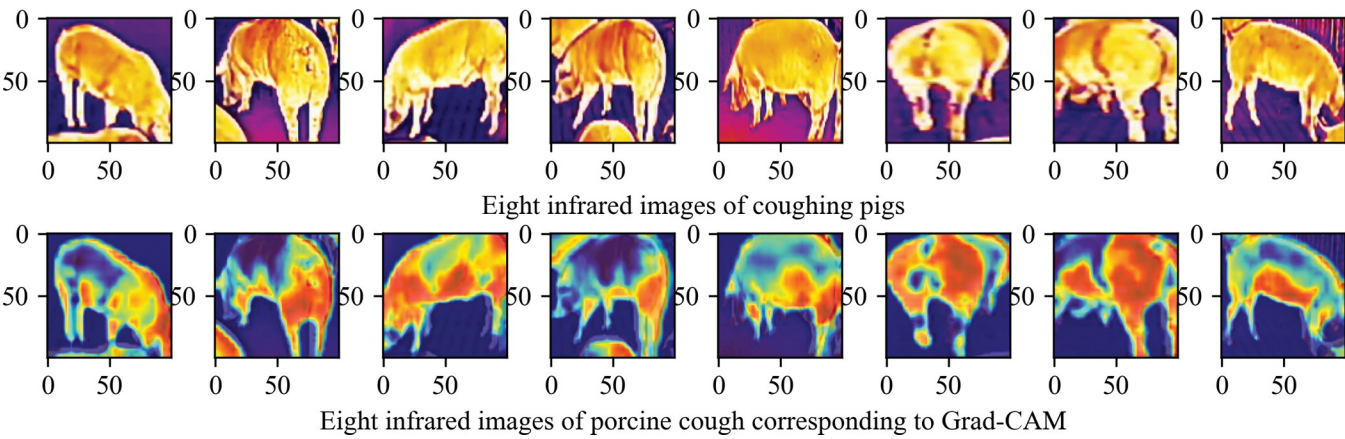


图14. 热成像网络 (ThermographicNet) 的梯度加权类激活映射。

<https://doi.org/10.1371/journal.pone.0297655.g014>

ThermographicNet model predominantly include the mouth, nose, ear, and groin of the pig. These key sites are entirely interpretable from the standpoint of animal physiology and pathology. Coughing induced by upper respiratory infections is primarily characterized by elevated temperatures in the ear canal, mouth, nose, and groin. This is reflected in the infrared data, where the infrared features of these specific parts are more prominent, and our feature extractor adeptly captures these features. Therefore, not only is it feasible to conduct research on automatic respiratory health detection using depth features, but it also holds physiological interpretability for pigs. This stands as one of the principal contributions of this paper. Furthermore, extracting physiological deep features from thermal infrared images may necessitate additional equipment, technical support, and a more complex data acquisition environment. This could involve procuring thermal infrared imaging equipment and implementing intricate image processing and analysis techniques, thereby increasing the research difficulty and cost and limiting its applicability and generalizability. Therefore, in the actual deployment, full consideration should be given to the integrated application of the existing monitoring and other systems in the farm.

The merits of our methods are primarily geared towards intensive farms, aiming for non-contact and stress-free automatic detection of the respiratory health status of pigs. The objective is to achieve early detection and treatment, significantly reducing the workload on farm staff, improving work efficiency, cutting farming costs, boosting profitability, and ultimately enhancing animal welfare. The experimental results substantiate the advantages and viability of our approach. However, our method is susceptible to interference from anomalies such as persistent high-frequency noise in the environment, unexpected intrusion of heat generators, and equipment shielding, among others. Consequently, we must employ additional measures and techniques in data preprocessing and deployment implementation, leading to a certain increase in costs and performance fluctuations. To address the impact of spatial distribution on data quality [60], we have employed a data acquisition system with a cloud-side-end architecture for data acquisition. Relevant equipment is strategically deployed in designated locations, following a standardized production deployment approach. This ensures continuous 24-hour data collection, covering a spectrum of acoustic variations throughout the entire production process. Moreover, it helps mitigate the impact of thermal attenuation resulting from distance and ambient space, effectively dealing with the challenges posed by spatial differences. While various factors may influence method performance, the disparity between experimental results and actual deployment applications is negligible. Recent advancements in biological and physiological sensing technologies have proven beneficial for improving livestock health. In the future, we intend to enhance swine respiratory health monitoring conditions by developing an all-in-one machine that can enhance the quality of audio and thermal images collected within the barn.

Conclusions

In this study, we propose a multimodal feature fusion approach that fuses acoustic source biometric features and thermal source physiological deep features in a multi-source classification framework to improve the recognition performance of cough in pigs. We enriched acoustic source biometric features by extracting thermal source deep physiological features from a shallow convolutional neural network. We utilized the complementary nature of different biological and physiological features to enhance cough recognition. Our study concludes that cough recognition using infrared images is effective in a swine barn environment, and the heterogeneous fusion method is more suitable for recognizing coughing behavior than the traditional acoustic homogeneous fusion. In the future, we can extend this study by applying other

ThermographicNet 模型主要提取的深度特征位点包括猪的口、鼻、耳和腹股沟部位。这些关键位点从动物生理学和病理学的角度来看完全可解释。由上呼吸道感染引起的咳嗽主要表现为耳道、口、鼻和腹股沟部位温度升高。这在红外数据中得以反映, 这些特定部位的红外特征更为突出, 而我们的特征提取器能够巧妙地捕捉这些特征。因此, 不仅可行使用深度特征进行自动呼吸健康检测的研究, 而且对猪而言具有生理可解释性。这构成了本文的一项主要贡献。此外, 从热红外图像中提取生理深度特征可能需要额外的设备、技术支持以及更复杂的数据采集环境。这可能涉及采购热红外成像设备并实施复杂的图像处理和分析技术, 从而增加研究难度和成本, 并限制其适用性和泛化性。因此, 在实际部署中, 应充分考虑农场现有监控和其他系统的综合应用。

我们的方法优势主要面向密集型农场, 旨在实现猪呼吸健康状况的非接触式、无压力自动检测。目标是实现早期发现和治疗, 显著减轻农场工作人员的工作量, 提高工作效率, 降低农场成本, 提升盈利能力, 并最终增强动物福利。实验结果证实了我们的方法的优势和可行性。然而, 我们的方法易受环境持续性高频噪声、热源意外侵入、设备屏蔽等异常干扰。因此, 我们必须在数据预处理和部署实施中采取额外的措施和技术, 导致成本和性能的某些增加。为解决空间分布对数据质量的影响 [60],我们采用了一种具有云端-边缘架构的数据采集系统。

相关设备按照标准化的生产部署方法, 在指定位置进行战略部署。这确保了连续的24小时数据采集, 覆盖整个生产过程中各种声学变化。此外, 它有助于减轻距离和环境空间导致的温度衰减影响, 有效应对空间差异带来的挑战。虽然各种因素可能影响方法性能, 但实验结果与实际部署应用之间的差异可以忽略不计。近年来, 生物和生理传感技术的进步已被证明对改善牲畜健康有益。未来, 我们计划通过开发一种能提升畜舍内采集的音频和热成像质量的集成机器, 来改善猪呼吸健康监测条件。

结论

在本研究中, 我们提出了一种多模态特征融合方法, 该方法在多源分类框架中将声源生物特征与热源深度生理特征融合, 以提升猪咳嗽的识别性能。我们通过从浅层卷积神经网络中提取热源深度生理特征, 丰富了声源生物特征。我们利用不同生物和生理特征的互补性来增强咳嗽识别。我们的研究表明, 使用红外图像进行咳嗽识别在猪舍环境中是有效的, 而异构融合方法比传统的声学同构融合更适合识别咳嗽行为。未来, 我们可以通过应用其他生物声学和生理红外样本来扩展这项研究, 这对于在田间环境下进行咳嗽分类、改善动物福利和实现智能畜牧业至关重要。

bioacoustic and physiological infrared samples for cough classification in field situations, which is essential for improving animal welfare and realizing smart animal husbandry.

Author Contributions

Conceptualization: Buyu Wang, Xiaoping An, Yuan Wang.

Data curation: Buyu Wang, Xiaoping An.

Formal analysis: Xiaoping An, Yuan Wang.

Funding acquisition: Buyu Wang, Jingwei Qi.

Investigation: Buyu Wang, Xiaoping An, Yuan Wang.

Methodology: Buyu Wang, Xiaoping An, Yuan Wang.

Project administration: Buyu Wang, Jingwei Qi.

Software: Buyu Wang.

Validation: Buyu Wang, Yuan Wang.

Visualization: Buyu Wang, Yuan Wang.

Writing – original draft: Buyu Wang.

Writing – review & editing: Buyu Wang.

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生物声学 and 生理红外样本进行咳嗽分类，这对于改善动物福利和实现智能畜牧业至关重要。

作者贡献

概念化: Buyu Wang, Xiaoping An, Yuan Wang.

数据管理: Buyu Wang, Xiaoping An.

形式分析: Xiaoping An, Yuan Wang.

资金获取: Buyu Wang, Jingwei Qi.

调查研究: Buyu Wang, Xiaoping An, Yuan Wang.

方法: 王买玉, 安晓平, 王媛。

项目管理: 王买玉, 齐景伟。

软件: 王买玉。

验证: 王买玉, 王媛。

可视化: 王买玉, 王媛。

撰写——初稿: 王买玉。

写作 – 审稿与编辑: Buyu Wang.

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